Precalculus II

An Original Ten-Week Course Reader

Dr. Lee MATH& 142

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1 Angles, Radians, and the Unit Circle

A turning wheel connects angle, distance, speed, and position. This chapter develops those connections from a coordinate sketch to a model of a rider's height over time. Work through each example with pencil and paper. Pause at each check before reading its explanation at the end of the chapter.

Learning objectives. Describe signed rotations; convert between degrees and radians; distinguish arc length, displacement, angular velocity, and tangential speed; derive exact unit-circle coordinates; and connect circular motion with periodic graphs and a height model.

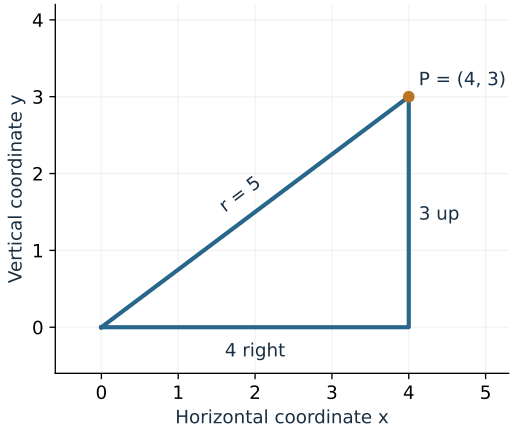
Online reading and interactive activities

Further reading: OpenStax, *Precalculus 2e*, 5.1 Angles and 5.2 Unit Circle: Sine and Cosine Functions.

1.1 Read a point before describing a turn

Imagine a marker on the rim of a turning wheel. We want to describe where the marker is, how far it has traveled, and how quickly it moves. These are different questions. Trigonometry connects them by treating an angle as a record of turning.

Begin with a coordinate plane. The origin is the point where the axes meet. In an ordered pair (x, y), the first number tells us horizontal position and the second tells us vertical position. Positive x means right; negative x means left. Positive y means above the horizontal axis; negative y means below it.



The coordinates describe two perpendicular moves. The distance from the origin is the hypotenuse.

A point and its distance from the origin

For $P = (4, 3)$, the Pythagorean theorem gives the distance r from the origin:

$$r^2 = 4^2 + 3^2 = 25, \qquad r = 5.$$

If we shrink this triangle until its hypotenuse is 1, every length is divided by 5. The new point is $(4/5, 3/5)$. It points in the same direction as $(4, 3)$, but lies on a circle of radius 1. This process is called *normalizing* the coordinates.

We will use fractions and square roots to describe some positions exactly. An expression such as $\sqrt{2}/2$ is an exact number. Its decimal approximation, about 0.707, is useful for estimating, but the rounded decimal loses some information. Both forms will appear in this module.

As you read, keep a sketch beside your calculations. A positive vertical coordinate should put the point above the horizontal axis; a distance should never be negative. These simple checks can catch many errors.

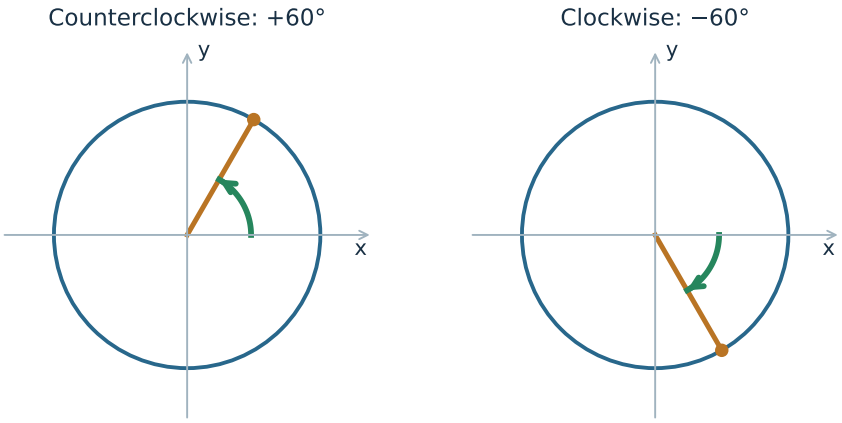
Check 1

A point is at $(-3, 4)$. Find its distance from the origin and divide both coordinates by that distance. Which signs should remain?

1.2 An angle records a signed rotation

An angle describes how a ray turns around its endpoint. A ray is a line that starts at one point and extends in one direction. In *standard position*, the endpoint is the origin and the starting ray points right along the positive x-axis. The final ray is called the *terminal ray*.

We use a sign to record the direction of turning. Counterclockwise turns are positive; clockwise turns are negative. A complete turn is 360°, a half-turn is 180°, and a quarter-turn is 90°. The Greek letter θ, pronounced "theta," often stands for an angle.



Begin at the positive horizontal axis. The sign tells you which way to turn.

Read the whole rotation

For 420°, turn counterclockwise through one full revolution and then another 60°:

$$420^\circ = 360^\circ + 60^\circ.$$

For -120°, turn clockwise through 120°. The terminal ray points down and left. You could reach that same ray by turning counterclockwise through 240°, but the two descriptions record different journeys.

A diagram usually shows the final direction more easily than it shows several complete turns. Do not

assume that an angle must be between zero and one revolution. A wheel may turn many times, and a negative angle is a perfectly valid description of motion.

Check 2

Where do $-90°$ and $450°$ end? Describe the direction and amount of turning for each.

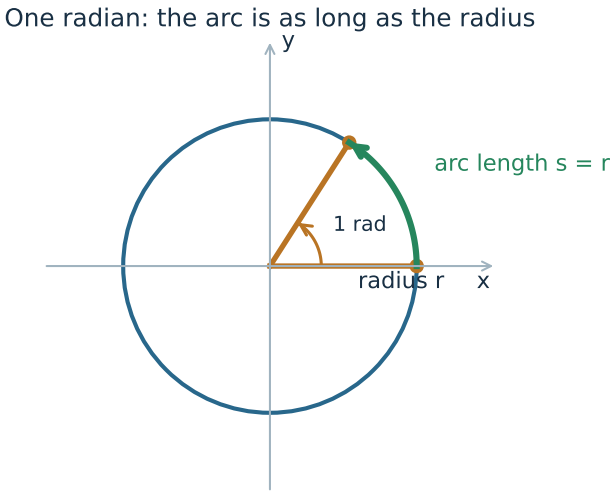
1.3 Radians compare an arc with a radius

Degrees divide a full turn into 360 equal pieces. Radians measure the same turning in a way tied directly to the circle. An *arc* is a curved part of a circle. For a central angle, the two rays meet at the circle's center and cut off an arc on its circumference.

Take the length of that arc and divide by the radius, using the same length units for both. The ratio is the angle's magnitude in radians:

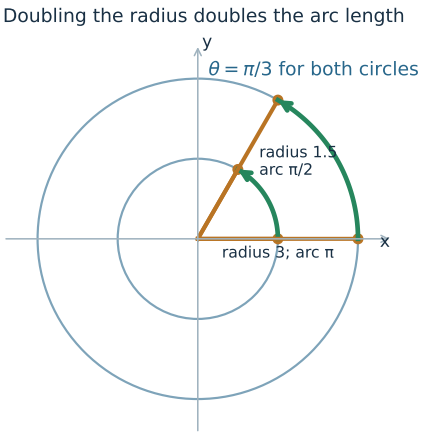
$$\boxed{\theta = \frac{s}{r}} \qquad \text{(radian measure)}.$$

An angle of 1 radian cuts off an arc whose length equals the radius. It is a little less than $60°$: about $57.3°$. A radian is an angle unit; it is not another name for π.



One radian means one radius-length of arc.

Why does the size of the circle not change the angle? If you double the radius while keeping the same two rays, the arc also doubles. The ratio s/r stays the same. The picture below shows this for an angle of $\pi/3$ radians.



Angle describes turning. Radius describes size. The ratio of arc length to radius separates these ideas.

Why one turn is 2π radians

A circle's circumference is $2\pi r$. For a complete turn, divide the full circumference by the radius:

$$\theta = \frac{2\pi r}{r} = 2\pi.$$

Thus $360^\circ = 2\pi$ radians and $180^\circ = \pi$ radians. The number π is about 3.14159. A half-turn has radian measure π because its arc has length πr.

For clockwise motion, use a negative angle. If s means ordinary arc length, it is nonnegative and s/r gives the angle's magnitude. A signed angle requires an orientation as well. We will keep signed arc length and distance traveled separate.

Check 3

An arc is 9 cm long on a circle of radius 6 cm. What is the angle's magnitude in radians? At the same angle on a circle of radius 12 cm, how long is the arc?

1.4 Convert units without changing the angle

The relation $180^\circ = \pi$ radians lets us write the same angle in either unit. To convert, multiply by a fraction that equals 1 and cancels the unwanted unit.

$$\text{Degrees to radians: multiply by } \frac{\pi \text{ rad}}{180^\circ}.$$

$$\text{Radians to degrees: multiply by } \frac{180^\circ}{\pi \text{ rad}}.$$

Degrees to radians: 240°

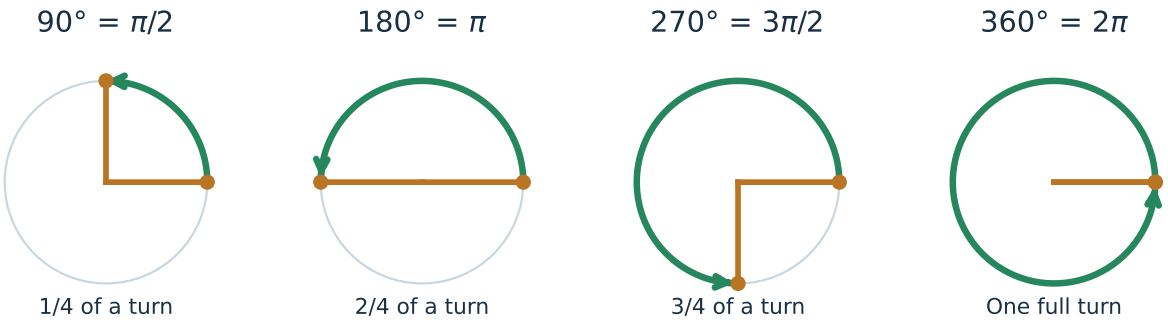
$$240^\circ \cdot \frac{\pi \text{ rad}}{180^\circ} = \frac{240\pi}{180} \text{ rad} = \frac{4\pi}{3} \text{ rad}.$$

The degree units cancel. Before calculating, we know that 240° is between a half-turn and a full turn. Our result, $4\pi/3$, is between π and 2π, as expected.

Radians to degrees: -5π/3

$$\left(-\frac{5\pi}{3} \text{ rad}\right) \frac{180^\circ}{\pi \text{ rad}} = -\frac{5(180^\circ)}{3} = -300^\circ.$$

The π factors and radian units cancel. Keep the negative sign: changing units does not change the direction of rotation.



A fraction of a full turn is the same fraction of 360° or of 2π radians.

Degrees	Radians	Fraction of one turn
30°	$\pi/6$	1/12
45°	$\pi/4$	1/8
60°	$\pi/3$	1/6
90°	$\pi/2$	1/4
180°	π	1/2
270°	$3\pi/2$	3/4
360°	2π	1

Leave an answer such as $4\pi/3$ exact unless a decimal is requested. To estimate it, use the calculator's π key and round at the end. Replacing π by 3.14 too early can add avoidable error.

Check the calculator's angle mode

In degree mode, $\sin(30) = 1/2$. In radian mode, the same turn is entered as $\sin(\pi/6) = 1/2$. Typing 30 in radian mode means 30 radians, a very different angle. Converting an angle and changing the calculator mode are separate actions.

Check 4

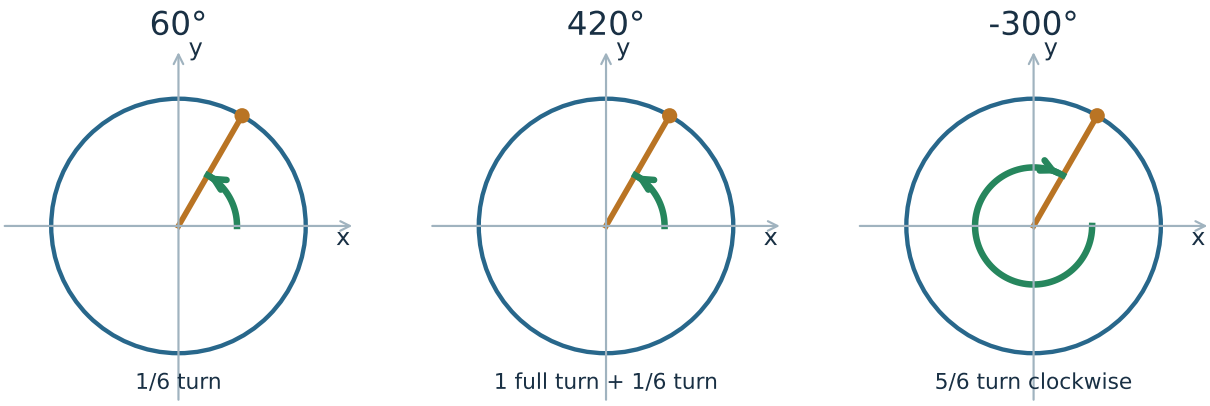
Convert $72°$ to radians and $7\pi/5$ radians to degrees. Check each result against one full turn.

1.5 Same final direction, different journeys

Two angles are *coterminal* when they have the same starting and terminal rays. Adding or subtracting a full turn changes the amount of rotation without changing the final direction.

$$\theta + 360°k \quad \text{or} \quad \theta + 2\pi k, \qquad k \in \mathbb{Z}.$$

The symbol $\mathbb{Z}$ means the integers: $\ldots, -2, -1, 0, 1, 2, \ldots$. Choose the formula that matches your angle unit. For example, $60°$, $420°$, and $-300°$ all end on the same ray.



A terminal ray does not tell us how many turns happened along the way.

Find a representative between 0 and 2π

Start with $-13\pi/4$. A full turn is $2\pi = 8\pi/4$, so add full turns using a common denominator:

$$-\frac{13\pi}{4} + \frac{8\pi}{4} = -\frac{5\pi}{4}, \qquad -\frac{5\pi}{4} + \frac{8\pi}{4} = \frac{3\pi}{4}.$$

Now $0 \le 3\pi/4 < 2\pi$. The terminal ray is in the upper-left part of the plane. We added two full turns; the original clockwise rotation was not erased from its physical history.

The interval $[0, 2\pi)$ includes 0 and excludes 2π. Those two angles point in the same direction, so

including both would repeat a representative. Similarly, use $[0°, 360°)$ for a single representative in degrees.

For the final point on a circle, a coterminal angle is enough. For distance traveled, use the actual rotation. A marker turning through 420° travels seven times as far as one turning through 60° on the same circle.

Check 5

Find the angle in $[0°, 360°)$ coterminal with 870°. How many full turns did you set aside?

1.6 Arc length, distance, and displacement

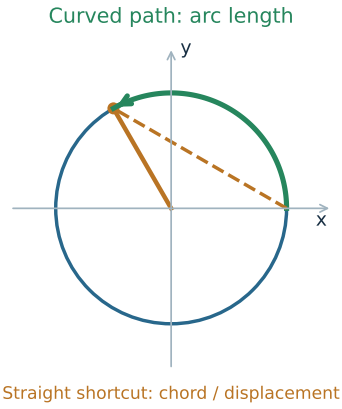
The equation $\theta = s/r$ can be rearranged to find arc length. When θ is a signed angle in radians, $r\theta$ gives a *directed* arc length:

$$s_{\text{directed}} = r\theta.$$

If the motion proceeds in one direction without reversing, the distance traveled is

$$\boxed{\text{distance} = r|\theta|}.$$

The absolute-value bars remove the sign. A clockwise journey still covers a positive distance. If motion reverses, calculate the distance for each part and add those distances. The final net angle alone will not tell you the whole distance.



Distance along the rim follows the arc. Displacement compares the starting and ending positions directly.

A clockwise turn on a radius-6 circle

A point turns clockwise through $120°$ on a circle of radius 6 cm. Convert first:

$$\theta = -120° \frac{\pi}{180°} = -\frac{2\pi}{3} \text{ rad.}$$

Then the directed arc is $6(-2\pi/3) = -4\pi$ cm, while the distance traveled is 4π cm, about 12.57 cm. The negative directed arc records clockwise orientation; it is not a negative distance.

Why radians are required

For a quarter-turn, $\theta = \pi/2$, so the arc is $r\pi/2$, one quarter of $2\pi r$. Putting 90 directly into $r\theta$ would give $90r$, which is much larger than the whole circumference. Convert degrees before multiplying.

Displacement is the change in position from the start to the finish. Its magnitude is the straight-line distance between the endpoints, not the arc length. After one complete revolution, a point has traveled a whole circumference but has zero displacement.

Check 6

A point on a circle of radius 2 m moves clockwise through half a turn. Find the signed angle, distance traveled, and magnitude of displacement.

1.7 Angular velocity and tangential speed

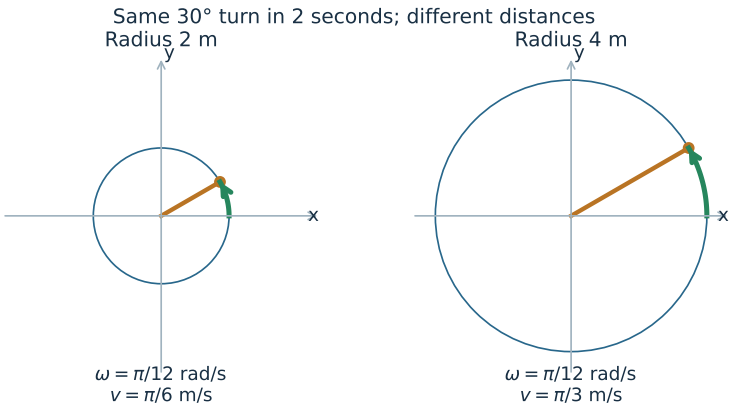
Angular velocity describes the rate at which the angle changes. Over a time interval of length Δt, the average angular velocity is

$$\omega = \frac{\Delta\theta}{\Delta t}.$$

The Greek letter ω is "omega," and Δ means "change in." With angles in radians and time in seconds, the units are radians per second, written rad/s. Counterclockwise angular velocity is positive; clockwise angular velocity is negative. Angular *speed* is the nonnegative magnitude $|\omega|$.

For uniform rotation, the rate is constant. A point farther from the center travels a longer arc during the same turn, so its tangential speed is greater:

$$\boxed{v = r|\omega|}.$$



The angular speed is shared by the whole wheel. Tangential speed depends on distance from the axis.

Convert revolutions per minute

A disk makes 6 counterclockwise revolutions per minute. A marker sits 0.5 m from the axis. Convert the angular rate one unit at a time:

$$6\frac{\text{rev}}{\text{min}} \cdot \frac{2\pi \text{ rad}}{1 \text{ rev}} \cdot \frac{1 \text{ min}}{60 \text{ s}} = \frac{\pi}{5} \text{ rad/s}.$$

The revolutions and minutes cancel. The marker's tangential speed is $v = 0.5(\pi/5) = \pi/10$ m/s, about 0.314 m/s. A marker twice as far from the axis would move twice as fast.

The *period* T is the time needed for one complete revolution. For uniform rotation, angular speed is $2\pi/T$. If the rotation is counterclockwise, $\omega = 2\pi/T$; if clockwise, $\omega = -2\pi/T$. A smaller period means a faster rotation.

Average speed and instantaneous speed agree here because the rotation rate is constant. If the wheel speeds up and slows down, one average rate describes only the chosen interval.

Check 7

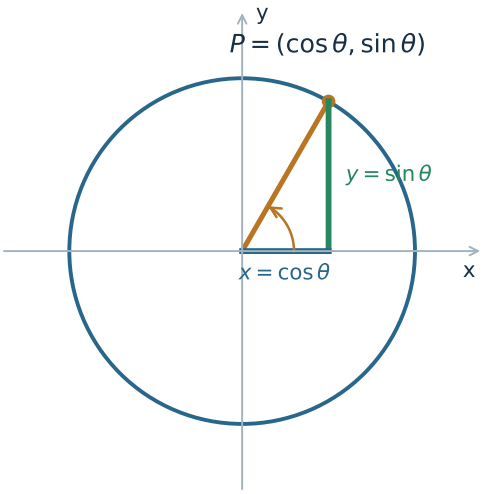
A wheel of radius 2 m makes one counterclockwise revolution every 20 seconds. Find its angular speed and tangential speed.

1.8 Sine and cosine are coordinates

The *unit circle* has center $(0, 0)$ and radius 1. Start at $(1, 0)$ and turn through angle θ. The coordinates of the resulting point define two functions:

$$\boxed{P = (\cos\theta, \sin\theta)}.$$

Cosine is the horizontal coordinate. Sine is the vertical coordinate. The expression $\sin\theta$ means "the sine of the angle θ," not "sin multiplied by theta." A function takes an input and assigns an output; here the input is an angle and the output is a coordinate.



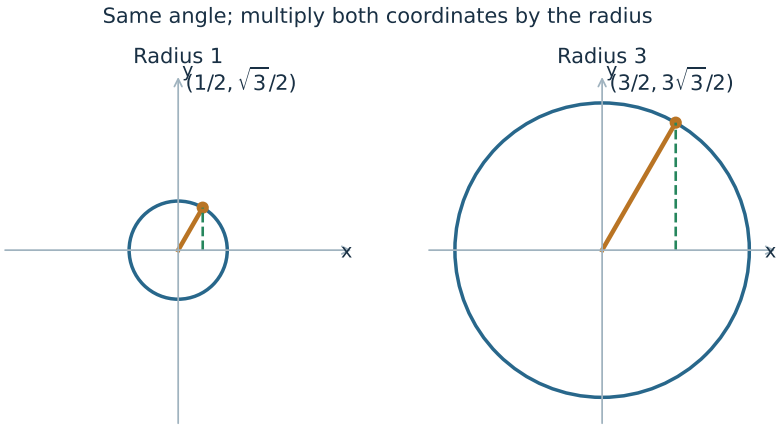
Keep the coordinate order: cosine first, sine second.

The familiar right-triangle ratios are built into this definition. In the first quadrant, the triangle has hypotenuse 1, adjacent side x, and opposite side y. Thus $\cos\theta = x/1 = x$ and $\sin\theta = y/1 = y$. The circle definition also handles angles beyond a right triangle, including negative angles and multiple turns.

Since every unit-circle point satisfies $x^2 + y^2 = 1$, its coordinates satisfy

$$\boxed{\cos^2\theta + \sin^2\theta = 1}.$$

The notation $\cos^2\theta$ means $(\cos\theta)^2$. Both coordinates lie between -1 and 1. These are useful checks on calculated values.



Changing the radius changes the physical coordinates, while sine and cosine of the angle stay the same.

Scale from a unit circle to a real wheel

If a point at angle θ has unit-circle coordinates $(3/5, 4/5)$, a point at that angle on a radius-5 circle has coordinates $(3, 4)$. In general, on a circle centered at the origin:

$$x = r\cos\theta, \qquad y = r\sin\theta.$$

These are physical coordinates. Dividing them by the radius recovers the unit-circle coordinates. The radius does not change the angle's sine or cosine.

Check 8

The unit-circle point at an angle is $(-3/5, -4/5)$. What is the corresponding point on a circle of radius 10 centered at the origin?

1.9 Derive exact values from two triangles

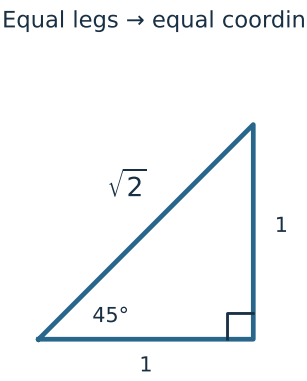
You do not need a separate rule for every special angle. Two right triangles supply the coordinate magnitudes for 30°, 45°, and 60°. The Pythagorean theorem explains where the square roots come from.

The 45° triangle

A right triangle with equal legs has two equal acute angles, each 45°. If each leg is 1, its hypotenuse is $\sqrt{1^2 + 1^2} = \sqrt{2}$. Divide every side by $\sqrt{2}$ to make the hypotenuse 1:

$$\cos 45^\circ = \sin 45^\circ = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}.$$

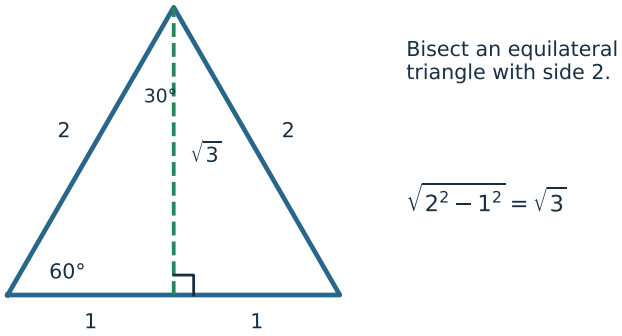
The last equality comes from multiplying top and bottom by $\sqrt{2}$. It changes the form of the number, not its value.



Equal legs give equal horizontal and vertical coordinates after normalization.

The 30° and 60° triangles

Cut an equilateral triangle of side 2 into two equal right triangles. Each has hypotenuse 2, short leg 1, and long leg $\sqrt{2^2 - 1^2} = \sqrt{3}$. The angles are 30°, 60°, and 90°.



Halving an equilateral triangle produces the ratio $1 : \sqrt{3} : 2$.

At 30°, the horizontal leg is $\sqrt{3}$ and the vertical leg is 1. Divide by the hypotenuse 2 to get $(\sqrt{3}/2, 1/2)$. At 60°, the roles of the two legs switch.

Angle	Radians	Cosine: horizontal	Sine: vertical
0°	0	1	0
30°	$\pi/6$	$\sqrt{3}/2$	$1/2$
45°	$\pi/4$	$\sqrt{2}/2$	$\sqrt{2}/2$
60°	$\pi/3$	$1/2$	$\sqrt{3}/2$
90°	$\pi/2$	0	1

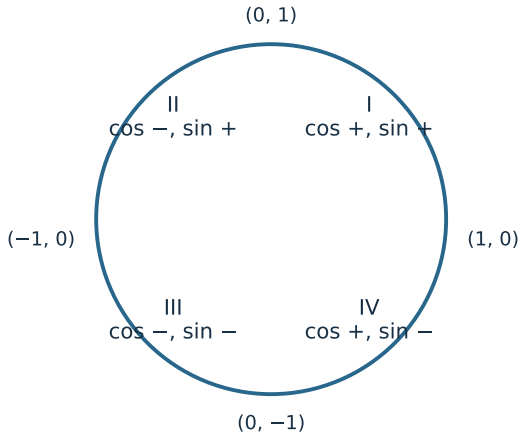
The axis points at 0° and 90° come directly from the circle. They do not require a right triangle with a zero-length leg.

Check 9

Why is $\sin 60°$ greater than $\sin 30°$? Check that the two coordinates at 45° satisfy the unit-circle equation.

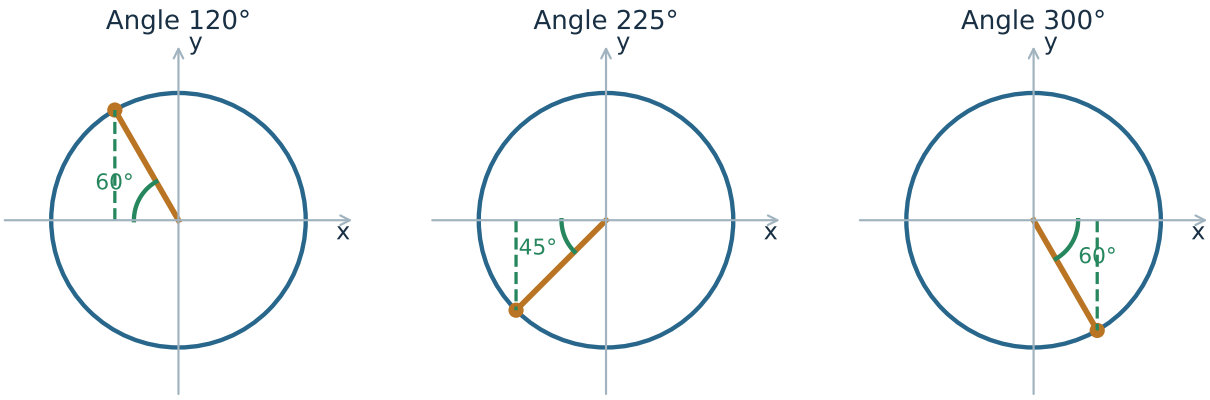
1.10 Use reference angles and quadrant signs

The axes divide the plane into four open regions called *quadrants*. Number them counterclockwise, beginning with the upper-right region. A point on an axis belongs to none of the open quadrants.



The signs follow from left/right and above/below. They are coordinate signs, not an extra rule to memorize.

For a terminal ray away from the axes, the *reference angle* is the acute angle between that ray and the horizontal axis. "Acute" means greater than $0°$ and less than $90°$. Always measure it to the horizontal axis, which may mean the negative x-axis.



Use the reference triangle for magnitudes, then the quadrant for signs.

Find the point at 120°

First locate the angle: $120°$ is in Quadrant II. The reference angle is $180° - 120° = 60°$. At $60°$, the coordinate magnitudes are $1/2$ and $\sqrt{3}/2$. Quadrant II lies left and up, so

$$(\cos 120°, \sin 120°) = \left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right).$$

The horizontal coordinate gets the minus sign. The vertical coordinate remains positive. Squaring and adding gives $1/4 + 3/4 = 1$, as it should.

When the original angle includes several turns, first find a coterminal angle in one revolution. Then locate its quadrant and reference angle. For a point on an axis, read the coordinates directly: $(1, 0)$, $(0, 1)$, $(-1, 0)$, or $(0, -1)$. No acute reference triangle is needed.

A useful routine is: locate the ray, find the acute reference angle, determine the two magnitudes, and assign each sign from the picture. Keep the order $(\cos\theta, \sin\theta)$.

Check 10

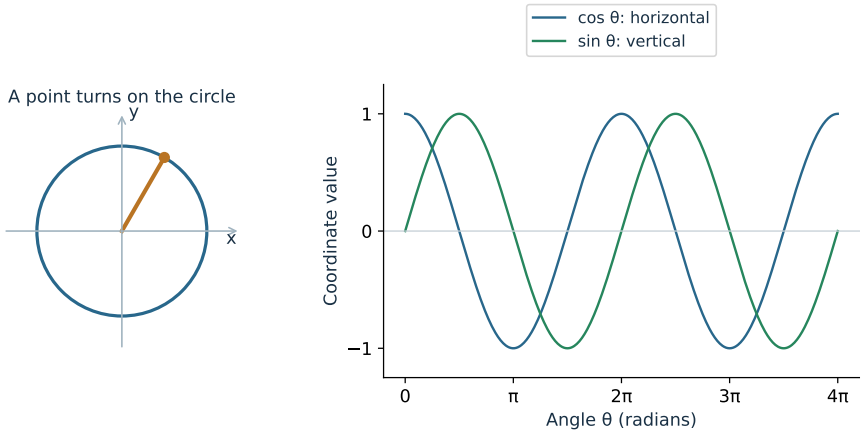
Find the unit-circle point at 300°. Does −60° give the same point? Explain.

1.11 A moving coordinate traces a wave

A point turning around a circle moves in two dimensions. We can also study one coordinate at a time. Make a graph whose horizontal axis is the angle θ, and whose vertical axis is the point's vertical coordinate. That graph is $y = \sin\theta$.

Be careful about the two pictures. On the circle, x and y describe physical position. On the graph of sine, the horizontal axis is the *angle input*; it is not the point's horizontal position.

Angle	Point on the circle	Cosine	Sine
0	$(1, 0)$	1	0
$\pi/2$	$(0, 1)$	0	1
π	$(-1, 0)$	-1	0
$3\pi/2$	$(0, -1)$	0	-1
2π	$(1, 0)$	1	0



One full turn produces one complete cycle of each coordinate graph.

Between these quarter turns, the point moves smoothly, so its coordinates change smoothly. Sine rises from 0 to 1, falls through 0 to -1, and returns to 0. Cosine begins at 1 because the starting point is $(1, 0)$.

A function is *periodic* if its values repeat after a fixed change in input. Adding a complete turn returns the point to exactly the same position:

$$\sin(\theta + 2\pi) = \sin\theta, \qquad \cos(\theta + 2\pi) = \cos\theta.$$

The period of each is 2π. A half-turn is generally not a period: it moves a point to the opposite side of the circle and reverses both coordinates. For instance, cosine is 1 at 0 and -1 at π.

Explore the connection

In Circle to Wave, stop at each quarter-turn and compare the point's height with the sine graph. Then show two full turns. The playback speed changes how quickly the demonstration runs; the graph against angle keeps the same shape. A graph against time requires a rotation rate, which we introduce next.

Check 11

Why does sine start at 0 while cosine starts at 1? How many complete cycles appear from 0 to 4π?

1.12 Build a model of height over time

To describe a turning wheel over time, connect two steps: time determines an angle, and the angle determines a position. For uniform rotation starting at angle zero,

$$\theta(t) = \omega t.$$

Here t is elapsed time. The notation $\theta(t)$ means the angle at time t. If ω is in rad/s and t is in seconds, their product is in radians.

For a wheel of radius r, centered horizontally at 0 and vertically at height c, with a rider starting at the rightmost point,

$$\boxed{x(t) = r\cos(\omega t), \qquad h(t) = c + r\sin(\omega t)}.$$

The sine term is vertical displacement from the center. Adding c gives height above ground. The radius stretches the unit-circle coordinate; it does not change the value of sine itself.

A radius-8 Ferris wheel

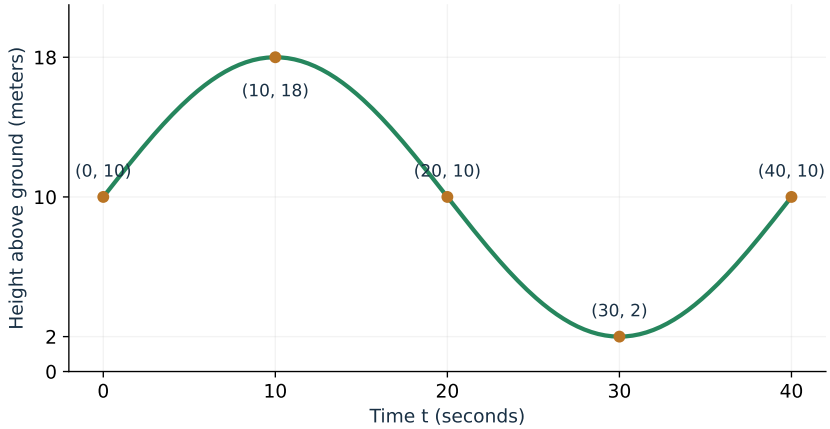
A wheel has radius 8 m, center height 10 m, and period 40 s. The rider starts at the rightmost point, and the wheel turns counterclockwise. First find the angular velocity:

$$\omega = \frac{2\pi}{40} = \frac{\pi}{20} \text{ rad/s}.$$

Next substitute the radius, center height, and angular rate:

$$x(t) = 8\cos\left(\frac{\pi t}{20}\right), \qquad h(t) = 10 + 8\sin\left(\frac{\pi t}{20}\right).$$

At $t = 10$, the angle is $\pi/2$, so the rider is at $(0, 18)$. This is the top: the 10 m center height plus the 8 m radius. At $t = 30$, the angle is $3\pi/2$, so the height is $10 - 8 = 2$ m.



The horizontal axis now measures seconds. The physical assumptions determine where the cycle begins and how long it takes.

Check a model against the situation before trusting a calculated value. The initial height should be 10 m, the height should stay between 2 and 18 m, and one full cycle should take 40 seconds. A negative height would signal a problem with this physical setup.

If a rider begins at another angle θ_0, use $\theta(t) = \theta_0 + \omega t$ inside the coordinate formulas. For clockwise rotation, use a negative ω. Starting position and direction matter; the radius and period alone do not determine a unique height formula.

This is the connection we will develop further: circular motion gives coordinate functions, and those functions let us model repeating behavior.

Check 12

In the example, what is the height at 20 seconds? If the radius became 12 m but the center stayed 10 m high, what would the model predict at the bottom?

1.13 Choose a relationship

Question	Relationship	What to check
How much turning?	$180^\circ = \pi$ rad	Keep the sign; match the unit.
Same terminal ray?	$\theta + 2\pi k$, $k \in \mathbb{Z}$	Use full turns, not half-turns.
How far along the rim?	$d = r\lvert\theta\rvert$	Radians; one direction without reversing.
How fast is the turning?	$\omega = \Delta\theta/\Delta t$	Signed rate, such as rad/s.
How fast is the rim point?	$v = r\lvert\omega\rvert$	Distance/time, such as m/s.
Where is the unit-circle point?	$(\cos\theta, \sin\theta)$	Horizontal first; check signs.
Where is the radius-r point?	$(r\cos\theta, r\sin\theta)$	Center at the origin.
Height on a wheel?	$h = c + r\sin(\omega t)$	Rightmost start, constant rotation rate.

1.14 Practice with explanation

Attempt each problem before consulting the explanations.

Try it. Sketch -120°, 240°, and 600°. Which terminal rays coincide? Justify the result using coterminal angles.

Try it. Convert 225° to radians and $-\pi/3$ radians to degrees, and sketch their terminal rays.

Try it. A Ferris wheel of radius 12 m makes one revolution every 40 s. Find its angular speed and a passenger's tangential speed.

Try it. A wheel of radius 0.6 m turns through 135° in 3 seconds. Find the arc length, average angular speed, and average tangential speed. Explain the units of all three answers.

Try it. Without a calculator, describe the point on the unit circle at 150° and at 330°. Which coordinates have the same absolute value? Which signs differ?

Try it. A different wheel has radius 6 meters, center height 7 meters, and period 30 seconds. Starting from the rightmost point, write its coordinate model and find the passenger's height after 7.5 seconds.

1.15 Review exercises

1. Convert -135° and 300° to radians.
2. Convert $11\pi/6$ and $-5\pi/4$ to degrees.
3. Find an angle in $[0, 2\pi)$ coterminal with $19\pi/6$.
4. A circle of radius 8 cm subtends an angle of $5\pi/12$. Find the arc length.
5. A turntable makes $33\frac{1}{3}$ revolutions per minute. Find angular speed in radians per second.
6. Explain why sine and cosine must lie between -1 and 1.
7. Find unit-circle coordinates at $2\pi/3$, $5\pi/4$, and $11\pi/6$.
8. A Ferris wheel has radius 10 m and center $(0, 12)$. Write a rider's coordinates after turning through θ from the rightmost point.
9. A point begins at $(1, 0)$ and turns through $-7\pi/6$ radians. Determine a coterminal angle in $[0, 2\pi)$ and give the exact point.
10. A wheel completes 9 revolutions in 2 minutes. Determine its angular speed in both radians per second and degrees per second.
11. A rider on a circle of radius 4 m moves through 75°. Find the arc length exactly in terms of π and approximately to two decimals.
12. Describe the coordinates at $3\pi/2$ and explain why a terminal point on an axis does not belong to any open quadrant.
13. For the Ferris-wheel model in the chapter, explain which physical quantity would change if the coefficient 8 in both coordinate equations became 12.

1.16 Explanations for the checks

1. The distance is $\sqrt{(-3)^2 + 4^2} = 5$. The normalized point is $(-3/5, 4/5)$. It stays left of the vertical axis and above the horizontal axis.

2. $-90°$ is one quarter-turn clockwise and ends on the negative y-axis. $450°$ is one full turn plus a quarter-turn counterclockwise and ends on the positive y-axis.

3. The angle is $9/6 = 1.5$ radians. Doubling the radius doubles the arc, so the new arc is 18 cm. The angle remains 1.5 radians.

4. $72(\pi/180) = 2\pi/5$ radians, one fifth of a turn. $(7\pi/5)(180/\pi) = 252°$, seven tenths of a turn.

5. $870° - 2(360°) = 150°$. We set aside two full turns. The original rotation was two turns plus $150°$.

6. The angle is $-\pi$ radians. Distance is $2|-\pi| = 2\pi$ m. The endpoints are opposite ends of a diameter, so the displacement has magnitude 4 m.

7. Angular speed is $2\pi/20 = \pi/10$ rad/s. Tangential speed is $2(\pi/10) = \pi/5$ m/s. The units distinguish turning rate from distance rate.

8. Multiply both coordinates by 10: $(-6, -8)$. Its distance from the origin is $\sqrt{36 + 64} = 10$, and both signs remain negative.

9. The point at $60°$ is higher: $\sqrt{3}/2 > 1/2$. At $45°$, $(\sqrt{2}/2)^2 + (\sqrt{2}/2)^2 = 1/2 + 1/2 = 1$.

10. The reference angle is $60°$, and Quadrant IV is right and down. The point is $(1/2, -\sqrt{3}/2)$. Since $-60° + 360° = 300°$, the angles are coterminal and give the same point.

11. The starting point is $(1, 0)$, so the horizontal coordinate is 1 and the vertical coordinate is 0. The interval from 0 to 4π contains two full turns and therefore two cycles.

12. At 20 seconds, $\theta = \pi$, so $h = 10$ m. A radius of 12 m with the same center would predict a bottom height of $10 - 12 = -2$ m. A wheel meant to stay above ground would need a higher center or a different physical setup.

1.17 Practice solutions

1. All three terminal rays coincide. $-120° + 360° = 240°$, and $600° - 360° = 240°$. They point down and left, although the signed rotations differ.

2. $225°(\pi/180°) = 5\pi/4$ radians, in Quadrant III. $(-\pi/3)(180°/\pi) = -60°$, in Quadrant IV.

3. One revolution is 2π radians, so angular speed is $2\pi/40 = \pi/20$ rad/s. Tangential speed is $12(\pi/20) = 3\pi/5$ m/s, about 1.885 m/s.

4. Convert $135° = 3\pi/4$ radians. Arc length is $0.6(3\pi/4) = 9\pi/20$ m. Average angular speed is $(3\pi/4)/3 = \pi/4$ rad/s. Average tangential speed is $(9\pi/20)/3 = 3\pi/20$ m/s, assuming the turn proceeds in one direction.

5. At $150°$, the point is $(-\sqrt{3}/2, 1/2)$. At $330°$, it is $(\sqrt{3}/2, -1/2)$. Corresponding coordinates have equal absolute values, and both signs are opposite.

6. The direction is not specified, so state an assumption. For counterclockwise rotation, $\omega = 2\pi/30 = \pi/15$, $x(t) = 6\cos(\pi t/15)$, and $h(t) = 7 + 6\sin(\pi t/15)$. At 7.5 s the angle is $\pi/2$, so the height is 13 m. Clockwise rotation would instead put the rider at height 1 m at that time.

1.18 Review solutions

1. $-135° = -3\pi/4$ radians; $300° = 5\pi/3$ radians. Multiply each degree value by $\pi/180$.

2. $11\pi/6 = 330°$; $-5\pi/4 = -225°$. Multiply each radian value by $180/\pi$.

3. $19\pi/6 - 2\pi = 19\pi/6 - 12\pi/6 = 7\pi/6$, which is in $[0, 2\pi)$.

4. $s = 8(5\pi/12) = 10\pi/3$ cm. The angle is already in radians.

5. $33\frac{1}{3} = 100/3$ rev/min. Multiply by 2π rad/rev and divide by 60 s/min to get $10\pi/9$ rad/s.

6. They are coordinates on a radius-1 circle centered at the origin. No such point can be farther than 1 unit horizontally or vertically from the origin.

7. At $2\pi/3$: $(-1/2, \sqrt{3}/2)$. At $5\pi/4$: $(-\sqrt{2}/2, -\sqrt{2}/2)$. At $11\pi/6$: $(\sqrt{3}/2, -1/2)$. Use the reference angle for magnitudes and the quadrant for signs.

8. $x = 10\cos\theta$, $y = 12 + 10\sin\theta$. The center adds 12 to the vertical coordinate; the radius multiplies both unit-circle coordinates.

9. $-7\pi/6 + 2\pi = 5\pi/6$. The point is $(-\sqrt{3}/2, 1/2)$, in Quadrant II.

10. In 120 seconds the wheel turns through 18π radians, or 3240°. Divide each by 120: the angular speed is $3\pi/20$ rad/s, or 27°/s.

11. $75° = 5\pi/12$ radians, so the arc is $4(5\pi/12) = 5\pi/3$ m, approximately 5.24 m.

12. The point is $(0, -1)$, on the negative y-axis. Open quadrants exclude their boundary axes, so it belongs to none of them.

13. The radius would increase from 8 to 12 m, increasing the amplitude of each coordinate function to 12. The period remains 40 s and the center height remains 10 m. The predicted height range becomes $[-2, 22]$ m; a wheel staying above ground would require a higher center.

2 The Six Trigonometric Functions

The six trigonometric functions are defined by coordinates on the unit circle and ratios of those coordinates. Their relationships determine their signs, domains, and graphs.

Let $P = (x, y)$ be the point at angle θ on the unit circle. Its coordinates are $x = \cos\theta$ and $y = \sin\theta$. The remaining four functions are formed from quotients and reciprocals of these coordinates.
Learning objectives. Define the six functions on the unit circle, calculate exact values in each quadrant, determine their domains, identify characteristic graph features, and apply trigonometric ratios to measurement problems.

2.1 From a point to six functions

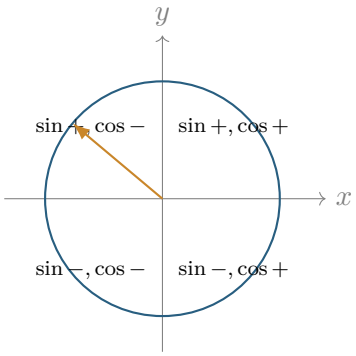
Last week we studied the unit-circle point $P = (x, y) = (\cos\theta, \sin\theta)$. The two coordinates determine four additional ratios whenever the denominators are nonzero:

$$\sin\theta = y, \quad \cos\theta = x, \quad \tan\theta = \frac{y}{x},$$
$$\csc\theta = \frac{1}{y}, \quad \sec\theta = \frac{1}{x}, \quad \cot\theta = \frac{x}{y}.$$

Thus the six functions encode related properties of a single point.

Definition 2.1. The *tangent* function is $\tan\theta = \sin\theta/\cos\theta$ where $\cos\theta \neq 0$. The functions csc, sec, cot are the reciprocals of sin, cos, tan respectively, wherever defined.

Example 2.1. At $\theta = \pi/3$ the circle point is $(1/2, \sqrt{3}/2)$. Thus $\tan\theta = \sqrt{3}$, $\sec\theta = 2$, $\csc\theta = 2/\sqrt{3} = 2\sqrt{3}/3$, and $\cot\theta = 1/\sqrt{3} = \sqrt{3}/3$.

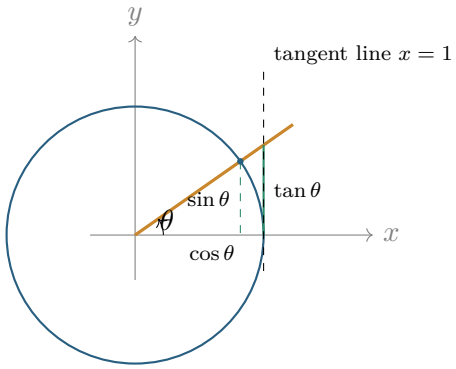


2.2 Tangent as a slope

For $\cos\theta \neq 0$, the line through the origin and $P = (\cos\theta, \sin\theta)$ has slope

$$m = \frac{\text{rise}}{\text{run}} = \frac{\sin\theta}{\cos\theta} = \tan\theta.$$

Tangent therefore gives the slope of the terminal line. Its magnitude increases without bound as the line approaches a vertical position, at which the slope is undefined.



The terminal line intersects $x = 1$ at $(1, \tan\theta)$, whenever $\cos\theta \neq 0$. Hence tangent also has a geometric interpretation as a directed length on a line tangent to the unit circle.

Example 2.2. At $30°$, the unit-circle coordinates are $(\sqrt{3}/2, 1/2)$. The slope of the corresponding ray is

$$\tan 30° = \frac{1/2}{\sqrt{3}/2} = \frac{\sqrt{3}}{3}.$$

This is also the slope of a line rising one unit for every $\sqrt{3}$ units of horizontal displacement.

2.3 Signs and the quadrants

The coordinate signs determine every trig-function sign. In quadrant I both coordinates are positive; in II, only sine is positive; in III, tangent is positive because both coordinates are negative; and in IV, only cosine is positive. Reciprocal functions have the same signs as their corresponding original functions.

Example 2.3. For $\theta = 5\pi/4$, the reference angle is $\pi/4$ and $P = (-\sqrt{2}/2, -\sqrt{2}/2)$. Therefore $\tan\theta = 1$, $\sec\theta = -\sqrt{2}$, and $\csc\theta = -\sqrt{2}$.

For a useful sign check, consider $150° = 5\pi/6$. The point is $(-\sqrt{3}/2, 1/2)$. Both sine and cosecant are positive, both cosine and secant are negative, and tangent and cotangent are negative:

$$\tan(5\pi/6) = -\frac{\sqrt{3}}{3}, \qquad \sec(5\pi/6) = -\frac{2\sqrt{3}}{3}, \qquad \csc(5\pi/6) = 2.$$

The signs of the six functions follow directly from the signs of sine and cosine.

	sin	cos	tan	csc	sec	cot
Quadrant I	+	+	+	+	+	+
Quadrant II	+	−	−	+	−	−
Quadrant III	−	−	+	−	−	+
Quadrant IV	−	+	−	−	+	−

Try it. The terminal point of an angle lies in quadrant IV and satisfies $\cos\theta = 3/5$. Determine $\sin\theta$, $\tan\theta$, and their reciprocals, showing how the quadrant tells you the signs.

2.4 Domains and undefined values

Division by zero is undefined. In particular, $\tan\theta$ and $\sec\theta$ do not exist where $\cos\theta = 0$, at $\pi/2 + k\pi$. Similarly $\cot\theta$ and $\csc\theta$ do not exist where $\sin\theta = 0$, at $k\pi$, for any integer k. Undefined values must be distinguished from zeros of a function.

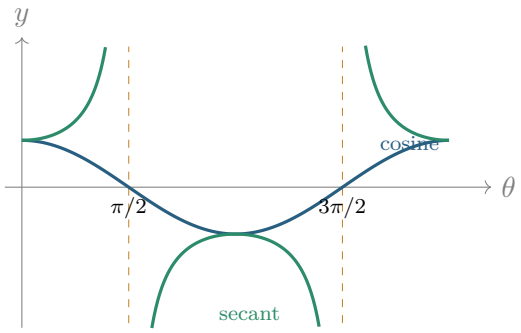
Try it. Find all six trig-function values at $\theta = 3\pi/2$, identifying every undefined value.

2.5 Zeros and vertical asymptotes

Near zeros of their denominators, trigonometric quotients may become arbitrarily large in magnitude. This behavior accounts for their vertical asymptotes.

For example, as θ approaches $\pi/2$ from the left, $\cos\theta$ approaches zero from above. Consequently $\sec\theta = 1/\cos\theta$ increases without bound. From the right, cosine is slightly negative, and secant becomes a large negative number. At $\pi/2$ itself, secant has no value.

The line $\theta = \pi/2$ is a vertical asymptote of both tangent and secant.



Only those portions of the secant graph within the displayed vertical range are shown. Its branches are separated by vertical asymptotes.

Try it. Describe the same phenomenon for $\csc\theta$ near $\theta = 0$. Explain why a zero of sine becomes an undefined value of cosecant.

2.6 Right-triangle ratios

For an acute angle θ in a right triangle, scale the triangle so its hypotenuse has length 1. The opposite and adjacent sides then give the coordinates $\sin\theta$ and $\cos\theta$. Restoring a general hypotenuse h gives

$$\sin\theta = \frac{\text{opposite}}{h}, \quad \cos\theta = \frac{\text{adjacent}}{h}, \quad \tan\theta = \frac{\text{opposite}}{\text{adjacent}}.$$

The unit-circle definitions extend the right-triangle ratios to arbitrary real angles.

Example 2.4. A ramp rises 1.5 meters over a horizontal run of 6 meters. Its angle of inclination satisfies $\tan\theta = 1.5/6 = 1/4$. Therefore $\theta = \arctan(1/4) \approx 14.0°$.

Writing o, a, and h for opposite side, adjacent side, and hypotenuse, respectively, gives

	opposite and hypotenuse	adjacent and hypotenuse	opposite and adjacent
direct	$\sin\theta = o/h$	$\cos\theta = a/h$	$\tan\theta = o/a$
reciprocal	$\csc\theta = h/o$	$\sec\theta = h/a$	$\cot\theta = a/o$

The six ratios describe the same triangle and agree with the unit-circle definitions for acute angles.

Example 2.5. A support cable stretches 13 meters from the ground to a point 12 meters above it. The horizontal run is $\sqrt{13^2 - 12^2} = 5$ meters. If θ is the angle with the ground, then

$$\sin\theta = \frac{12}{13}, \quad \cos\theta = \frac{5}{13}, \quad \tan\theta = \frac{12}{5}, \quad \csc\theta = \frac{13}{12}, \quad \sec\theta = \frac{13}{5}, \quad \cot\theta = \frac{5}{12}.$$

Each quotient expresses a relation between lengths of the same triangle.

2.7 Exact values at special angles

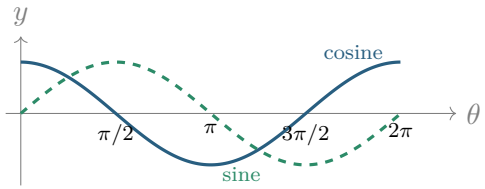
An equilateral triangle of side 2 splits into two right triangles with sides $1, \sqrt{3}, 2$. This produces $\sin 30° = 1/2$ and $\cos 30° = \sqrt{3}/2$. A square of side 1 has diagonal $\sqrt{2}$, producing $\sin 45° = \cos 45° = \sqrt{2}/2$.



2.8 Graphs of the trigonometric functions

On a trigonometric graph, the independent variable is an angle and the dependent variable is a coordinate or ratio at that angle. Such a graph is distinct from the circular trajectory of the associated point.

Sine and cosine remain between -1 and 1. Both repeat after a full rotation, so their fundamental period is 2π radians, or 360°. Tangent and cotangent repeat after half a turn, so their fundamental period is π radians, or 180°. Secant and cosecant share the 2π period of their reciprocal partners.



The zeros, excluded inputs, and fundamental period determine important features of a trigonometric graph.

Example 2.6. Rotation by π changes the signs of both unit-circle coordinates. Thus

$$\frac{\sin(\theta + \pi)}{\cos(\theta + \pi)} = \frac{-\sin\theta}{-\cos\theta} = \tan\theta.$$

Therefore tangent has period π on its domain.

The accompanying Six Trigonometric Graphs and Reciprocal Trigonometric Graphs demonstrations illustrate these functions and their asymptotes.

Try it. Sketch $y = \tan x$ and $y = \cot x$ on $(0, \pi)$, marking their zeros and asymptotes. How do the two plots differ at the interval's endpoints?

2.9 Applications: orientation and measurement

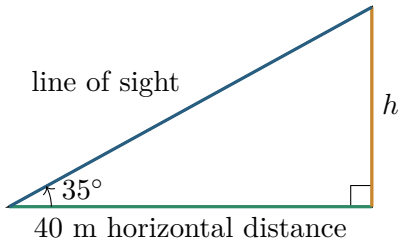
Suppose an observer stands 40 m from the foot of a vertical tower. The angle of elevation to its top is 35°. The tower's height h satisfies

$$\tan 35^\circ = \frac{h}{40}, \qquad h = 40 \tan 35^\circ \approx 28.0 \text{ m}.$$

Here the tangent function links an angle to a ratio of lengths. Always sketch the physical situation and label which distances are opposite and adjacent before calculating.

2.10 Measurement with trigonometric ratios

The appropriate trigonometric ratio depends on the available lengths. Tangent relates height to horizontal distance; sine relates height to line-of-sight distance.



Worked example: height measurement. An observer is 20 meters from the base of a tree and measures an angle of elevation of 40° to its top. If the observer's eyes are 1.6 meters above ground, the elevation of the treetop above eye level is

$$h_{\text{above eyes}} = 20 \tan 40^\circ \approx 16.8 \text{ m}.$$

The tree height is therefore about $16.8 + 1.6 = 18.4$ meters. The observer's eye height must be added to determine the height above ground.

Try it. A rescue drone is observed at an elevation angle of 25° from an observer whose eye level is 1.7 meters above ground. The point directly beneath the drone is 90 meters away horizontally. Estimate the drone's altitude.

2.11 Exercises

1. At $\theta = 2\pi/3$, find all six functions exactly.
2. At $\theta = 7\pi/4$, find $\tan, \cot, \sec, \csc$.
3. Explain geometrically why $\tan(\pi/2)$ is undefined.
4. For which angles in $[0, 2\pi)$ is $\csc\theta$ undefined?
5. A right triangle has opposite side 5 and adjacent side 12. Find all six trig ratios exactly.
6. A ladder leans against a wall, reaching 8 m high. Its foot is 3 m from the wall. Find the angle with the ground.
7. A drone is viewed at an elevation angle of 28° from a point 75 m horizontally from the point below it. Estimate its altitude.

8. Show that $\sin^2\theta + \cos^2\theta = 1$ implies $1 + \tan^2\theta = \sec^2\theta$ wherever $\cos\theta \neq 0$.
9. Find the exact values of all six trig functions at $11\pi/6$.
10. Describe the zeros and vertical asymptotes of $y = \tan x$ on $[-2\pi, 2\pi]$.
11. Sketch one period of $y = \sec x$, labeling asymptotes and its lowest positive and highest negative values.
12. Explain why neither $\csc x$ nor $\sec x$ can equal $1/2$.
13. If $\cos\theta = -12/13$ and θ lies in quadrant II, determine all other trig values exactly.
14. Show using the unit circle that $\tan(\theta + \pi) = \tan\theta$ wherever both sides are defined.
15. A point on a rotating wheel has unit-circle coordinates $(0, -1)$. Identify the angles in $[0, 2\pi)$ and all four undefined or zero-valued functions.

3 Periodic Functions and Their Graphs

Periodic functions describe quantities that repeat as their input increases. Trigonometric functions provide fundamental examples: their periodicity arises from rotation, and their graphs can be transformed to represent a wide range of oscillatory processes.

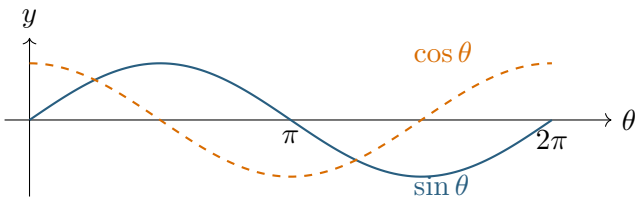
Learning objectives. Identify periods, amplitudes, midlines, and phase shifts; construct transformed sine and cosine graphs; distinguish angular frequency from frequency; and determine sinusoidal models from geometric or measured information.

3.1 Circular motion and periodicity

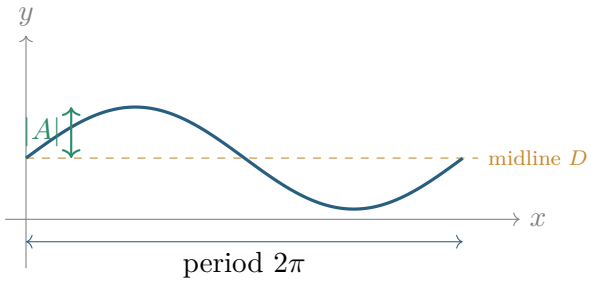
For a point traveling around the unit circle with angle θ, its vertical coordinate is $y = \sin\theta$ and its horizontal coordinate is $x = \cos\theta$. Each revolution restores the same coordinates. Hence

$$\sin(\theta + 2\pi) = \sin\theta, \qquad \cos(\theta + 2\pi) = \cos\theta.$$

A function f is *periodic* if there exists $T > 0$ such that $f(x + T) = f(x)$ for every x in its domain. When a least positive such T exists, it is the *fundamental period*. The fundamental periods of sine and cosine are both 2π. Moreover, $\sin 0 = 0$ and $\cos 0 = 1$.



3.2 Amplitude and the midline



3.3 Vertical transformations

Consider $y = A\sin x + D$. Its midline is $y = D$, its amplitude is $|A|$, and its range is $[D - |A|, D + |A|]$. A negative A reflects the wave across the midline.

Example 3.1. The function $y = 3\sin x - 2$ has amplitude 3, midline $y = -2$, maximum 1, and minimum -5.

For a sinusoid with maximum M and minimum m, the amplitude and midline are

$$|A| = \frac{M - m}{2}, \qquad D = \frac{M + m}{2}.$$

These relations determine the vertical parameters without knowing the phase or period.

Example 3.2. A periodically varying temperature has maximum 26°C and minimum 14°C. Its midline is 20°C and its amplitude is 6°C. These values do not determine the period or phase.

Try it. A sinusoidal signal ranges from -7 to 5. Determine its amplitude, midline, and range, and sketch one possible cycle.

3.4 Period and horizontal scaling

For $y = \sin(Bx)$ with $B \neq 0$, one full cycle occurs when Bx changes by 2π; the period is $T = 2\pi/|B|$.

Example 3.3. The graph of $y = \cos(4x)$ repeats every $\pi/2$ units of input.

When x denotes time measured in seconds, the quantity $|B|$ is angular frequency in radians per second, and ordinary frequency is $f = 1/T = |B|/(2\pi)$ cycles per second.

A phase change of 2π radians corresponds to one cycle. If the argument of a sinusoidal function is $\omega t + \phi$, then its period is $T = 2\pi/|\omega|$, and its frequency is

$$f = \frac{1}{T} = \frac{|\omega|}{2\pi}.$$

Frequency is measured in cycles per unit time, while angular frequency is measured in radians per unit time. These units describe the same repetition rate in different ways.

Example 3.4. A mechanism rotates at 90 revolutions per minute. Its frequency is 1.5 cycles per second, and its angular speed is $2\pi(1.5) = 3\pi$ radians per second. Its period is 2/3 second.

3.5 Phase shift

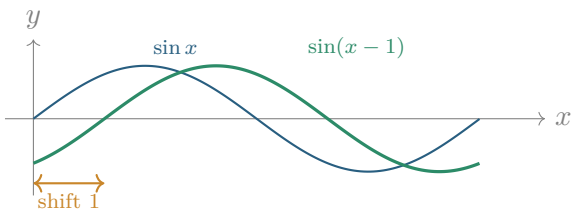
A useful general form is

$$\boxed{y = A\sin(B(x - C)) + D}.$$

Here C is the horizontal shift, D the midline, $|A|$ the amplitude, and $2\pi/|B|$ the period. We should distinguish $B(x - C)$ from $Bx - C$: factoring first prevents an incorrect phase shift.

Example 3.5. For $y = 2\sin(3x - \pi) + 1 = 2\sin(3(x - \pi/3)) + 1$, the period is $2\pi/3$, the shift is $\pi/3$ to the right, the amplitude is 2, and the midline is 1.

The parameterization assumes $B \neq 0$. In factored form, the phase shift is C. In the equivalent form $A\sin(Bx + \phi) + D$, the same shift is $C = -\phi/B$. The sign of B may be taken positive in a standard sine model by modifying the sign of A.



3.6 Five landmark points

Rather than sketching a sinusoid by guesswork, divide one period into four equal intervals. For $A > 0$, sine begins on the midline, rises to a maximum after one quarter-period, returns to the midline, reaches a minimum, then returns to the midline. A negative amplitude reverses the rises and falls.

Try it. Find five landmark points of $y = -2\cos(\pi(x-1)) + 3$, showing one entire period.

For $y = A\sin(B(x-C)) + D$ with $B > 0$, write $T = 2\pi/B$. One cycle is determined by the following five points:

x	C	$C+T/4$	$C+T/2$	$C+3T/4$	$C+T$
y	D	$D+A$	D	$D-A$	D

For cosine, the corresponding outputs are $D+A, D, D-A, D, D+A$. These points determine the positions of maxima, minima, and midline crossings; the smooth sinusoidal curve passes through them.

Example 3.6. For $y = -2\cos(\pi(x-1)) + 3$, the period is $T = 2$. Beginning at $x = 1$, the five landmark points are

$$(1,1), \quad (3/2,3), \quad (2,5), \quad (5/2,3), \quad (3,1).$$

The negative coefficient places the minimum at the start of the cycle.

3.7 Models of repeating motion

A Ferris wheel has radius 9 m, its center is 11 m above ground, and it makes one revolution every 30 s. If a passenger begins at the rightmost position and travels counterclockwise, their height is

$$h(t) = 11 + 9\sin\left(\frac{2\pi}{30}t\right).$$

Its amplitude is 9 m, midline is 11 m, and period is 30 s. A sinusoidal model can describe a wheel, a spring, a tide, or a musical tone, provided its parameters actually correspond to the chosen physical quantities.

3.8 Constructing a model from measurements

Suppose a buoy reaches a maximum height of 5 meters and a minimum height of 1 meter. Successive maxima are separated by 8 seconds. Taking $t = 0$ at a maximum yields the model

$$h(t) = 3 + 2\cos\left(\frac{\pi t}{4}\right).$$

The midline is 3 meters, the amplitude is 2 meters, and the angular frequency is $\pi/4$ radians per second. If the time origin were chosen at an upward midline crossing, a sine model would be more convenient. The physical motion is unchanged by this choice of representation.

Try it. A tide measurement varies between 0.8 m and 3.6 m and repeats every 12 hours. Write a cosine model assuming that the maximum occurs at $t = 2$ hours. State the meaning and units of every parameter.

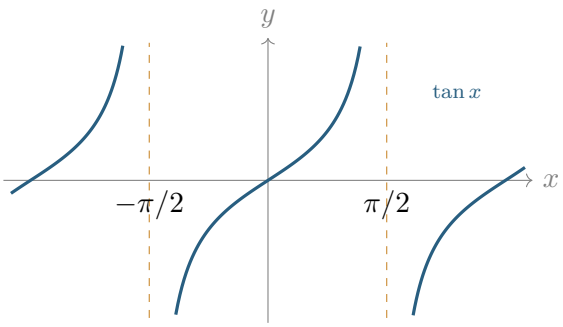
3.9 Tangent and reciprocal graphs

The tangent function repeats after π, not 2π, because both sine and cosine change sign after a half turn and their ratio remains the same:

$$\tan(x+\pi) = \tan x.$$

Tangent has vertical asymptotes at $x = \pi/2 + k\pi$. The secant and cosecant graphs arise as reciprocals of cosine and sine; their breaks occur wherever the respective denominator is zero.

Because $\tan x = \sin x/\cos x$, its zeros occur at $x = k\pi$ and its vertical asymptotes at $x = \pi/2 + k\pi$. Similarly, $\cot x = \cos x/\sin x$ is undefined at integer multiples of π. The graphs of secant and cosecant lie outside the strip $-1 < y < 1$ because their absolute values are at least 1 where defined.



3.10 Exercises

1. Find the amplitude, midline, period, and range of $y = -4\sin(2x) + 1$.
2. Rewrite $y = 3\cos(5x - \pi) + 2$ in shifted form and identify the phase shift.
3. Give five landmark points for $y = 2\sin(x - \pi/2)$.
4. Explain why $y = \sin(2x)$ completes two periods on $[0, 2\pi]$.
5. A wheel of radius 6 m rotates once every 24 seconds. Its center is 8 m above the ground. Model the rider's height when starting at the rightmost point.
6. Identify the vertical asymptotes of $y = \tan(2x)$ on $[0, 2\pi)$.
7. Sketch one period of $y = 2\sec x$ and indicate all discontinuities.
8. Explain how you can determine amplitude and midline directly from observed maximum and minimum values.
9. Determine a sinusoidal model with maximum 8, minimum -2, period 10, and maximum at $x = 3$.
10. Explain why $A\sin(B(x-C))+D$ is constant when $A = 0$, and why the usual fundamental-period formula does not apply.
11. A motor oscillates at 2.5 cycles per second. Determine its angular frequency and period.
12. Give the domain of $y = \tan(3x)$ and identify two consecutive vertical asymptotes.
13. Determine the amplitude and midline from a sinusoid with maximum 12.4 and minimum 7.6.

4 Inverse Trigonometry and Identities

Inverse trigonometric functions recover principal angles from trigonometric values. Their definitions require restrictions of the original domains. Trigonometric identities, in turn, express relationships among functions of the same angle and provide methods for simplifying and evaluating expressions.

Learning objectives. Determine the domains and ranges of the principal inverse trigonometric functions; evaluate and compose inverse and trigonometric functions; recover exact values from right triangles; prove identities using fundamental relations; and derive and apply angle-addition and double-angle formulas.

4.1 Inverse functions and principal branches

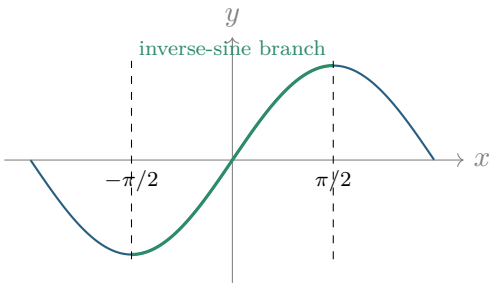
The notation $\sin^{-1} x$ commonly denotes $\arcsin x$, the inverse sine function. It must not be confused with $\csc x = 1/\sin x$. The inverse returns an angle from a prescribed trigonometric value, whereas the reciprocal returns a multiplicative inverse of that value.

A periodic function is not one-to-one on its full domain and therefore does not admit a global inverse function. To define the principal inverse sine, sine is restricted to $[-\pi/2, \pi/2]$, where it is strictly increasing and maps onto $[-1, 1]$. Similarly, cosine is restricted to $[0, \pi]$ and tangent to $(-\pi/2, \pi/2)$.

$f(x)$	Domain	Range
$\arcsin x$	$[-1, 1]$	$[-\pi/2, \pi/2]$
$\arccos x$	$[-1, 1]$	$[0, \pi]$
$\arctan x$	$\mathbb{R}$	$(-\pi/2, \pi/2)$

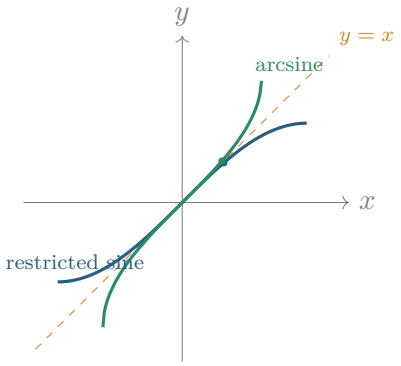
Example 4.1. $\arcsin(1/2) = \pi/6$, $\arccos(-1/2) = 2\pi/3$, and $\arctan(-1) = -\pi/4$.

Example 4.2. $\arcsin(\sin(5\pi/6)) = \pi/6$, not $5\pi/6$, because the inverse reports its principal value.



4.2 Reflection across the diagonal

The graphs of an invertible function and its inverse are reflections across the line $y = x$: if (a, b) lies on the graph of f, then (b, a) lies on the graph of f^{-1}. For inverse sine, only the restricted sine branch is reflected, rather than the entire periodic sine graph.



The reflected domains and ranges exchange roles. In particular, $\arcsin : [-1,1] \to [-\pi/2, \pi/2]$ whereas the restricted sine maps $[-\pi/2, \pi/2]$ onto $[-1,1]$.

Try it. List three points on $y = \cos x$ for $0 \le x \le \pi$, and interchange their coordinates to obtain points on $y = \arccos x$.

4.3 Compositions and triangle geometry

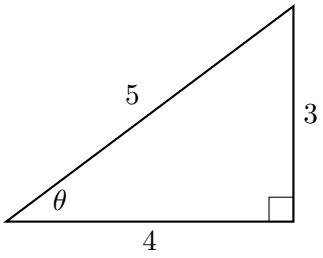
The identity $\sin(\arcsin x) = x$ holds for $-1 \le x \le 1$, but $\arcsin(\sin\theta) = \theta$ only for θ in the principal range. Suppose $\theta = \arctan(3/4)$. Then a right triangle has legs $3, 4$ and hypotenuse 5, so

$$\sin(\arctan(3/4)) = \frac{3}{5}.$$

For the inverse compositions, the domain restrictions determine which cancellations are valid:

$$\arcsin(\sin x) = x \quad \text{for } -\pi/2 \le x \le \pi/2, \qquad \sin(\arcsin u) = u \quad \text{for } -1 \le u \le 1.$$

The first identity does not hold for arbitrary real x, since the arcsine output is restricted to its principal range. Analogous statements hold for cosine and tangent on their chosen branches.



$\theta = \arctan(3/4)$

$\sin\theta = 3/5$

Example 4.3. To evaluate $\cos(\arcsin(-5/13))$, let $\theta = \arcsin(-5/13)$. Since $\theta \in [-\pi/2, \pi/2]$, its cosine is nonnegative. Thus

$$\cos\theta = \sqrt{1 - \sin^2\theta} = \sqrt{1 - \frac{25}{169}} = \frac{12}{13}.$$

The principal range determines the sign of the square root.

Try it. Evaluate $\sin(\arccos(-3/5))$ exactly. Explain the sign of the result using the principal range of arccosine.

4.4 Identities and their domains

From the unit-circle equation we obtain

$$\boxed{\sin^2 x + \cos^2 x = 1}.$$

Dividing by $\cos^2 x$ (only where $\cos x \neq 0$) gives $1+\tan^2 x = \sec^2 x$. Dividing by $\sin^2 x$ gives $1+\cot^2 x = \csc^2 x$ where $\sin x \neq 0$.

Example 4.4. For $\sin x \neq 0$,

$$\frac{1-\cos^2 x}{\sin x} = \frac{\sin^2 x}{\sin x} = \sin x.$$

This is an algebraic proof on the shared domain, not a numerical check at a few inputs.

4.5 Symmetry identities

The unit circle is symmetric about both coordinate axes. Reflection across the horizontal axis changes the sign of the vertical coordinate without changing the horizontal coordinate. Consequently,

$$\cos(-x) = \cos x, \qquad \sin(-x) = -\sin x.$$

The cosine function is *even*, while sine is *odd*. Taking quotients gives $\tan(-x) = -\tan x$ wherever tangent is defined.

A half-turn changes both coordinate signs:

$$\cos(x+\pi) = -\cos x, \qquad \sin(x+\pi) = -\sin x.$$

These relations explain the π-periodicity of tangent.

Try it. Use the unit-circle symmetries to evaluate $\sin(-7\pi/6)$ and $\cos(-7\pi/6)$ without a calculator.

4.6 Angle addition and double angles

Two successive plane rotations produce the sum identities

$$\cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta,$$
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta.$$

Set $\beta = -\gamma$ for difference identities and $\beta = \alpha$ for double-angle identities:

$$\sin(2x) = 2\sin x\cos x, \qquad \cos(2x) = \cos^2 x - \sin^2 x = 1 - 2\sin^2 x.$$

Example 4.5. $\sin(75^\circ) = \sin(45^\circ + 30^\circ) = (\sqrt{6}+\sqrt{2})/4$.

4.7 A geometric derivation of the addition formulas

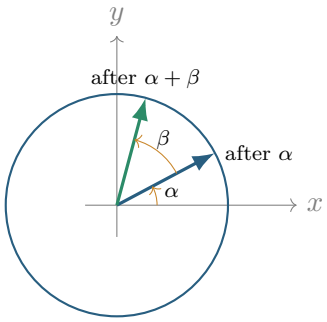
Rotation through an angle α sends a point (x, y) to

$$\begin{pmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}.$$

In particular, apply this rotation to the unit-circle point $(\cos\beta, \sin\beta)$. The result must be the point $(\cos(\alpha+\beta), \sin(\alpha+\beta))$, since successive rotations add their angles. Matrix multiplication gives

$$\cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta,$$
$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta.$$

Thus the sum formulas express the geometry of composing rotations, rather than independent algebraic rules.



4.8 Double-angle and half-angle consequences

Setting $\alpha = \beta = x$ gives

$$\sin 2x = 2\sin x \cos x, \qquad \cos 2x = \cos^2 x - \sin^2 x = 1 - 2\sin^2 x = 2\cos^2 x - 1.$$

Rearranging the cosine identity produces the power-reduction relations

$$\sin^2 x = \frac{1 - \cos 2x}{2}, \qquad \cos^2 x = \frac{1 + \cos 2x}{2}.$$

These expressions are particularly useful when simplifying squared trigonometric functions.

Example 4.6. The exact value of $\cos 15^\circ$ follows by using $15^\circ = 45^\circ - 30^\circ$:

$$\cos 15^\circ = \cos 45^\circ \cos 30^\circ + \sin 45^\circ \sin 30^\circ = \frac{\sqrt{6} + \sqrt{2}}{4}.$$

Try it. Use the addition identities to find $\sin 105^\circ$ exactly. Then verify the sign from the quadrant containing 105°.

4.9 Applications: measured angles

An observation of a tower from 40 m away gives $\tan\theta = 3/4$. The elevation angle is $\arctan(3/4) \approx 36.9^\circ$, while the height is exactly $40(3/4) = 30$ m. Inverse functions recover angles; ratios can often recover distances directly.

4.10 Recovering angles in measurement problems

The inverse function identifies an angle from a ratio, provided the physical geometry determines an appropriate branch. If a right triangle has adjacent side 40 and opposite side 30, its acute angle satisfies

$$\theta = \arctan(30/40) = \arctan(3/4) \approx 36.87^\circ.$$

The same geometry gives the opposite side directly from $h = 40\tan\theta$, when the angle is already known. In a measurement context, the acute-angle restriction is supplied by the geometry of the triangle.

4.11 Exercises

1. Evaluate exactly $\arcsin(-\sqrt{2}/2)$ and $\arctan(\sqrt{3})$.

2. Explain why $\arcsin(\sin(3\pi/4)) \neq 3\pi/4$.

3. Find $\cos(\arcsin(5/13))$ with a triangle.
4. Find $\sin(\arctan(-2))$ exactly, including its sign.
5. Prove $(\sec^2 x - 1)/\tan x = \tan x$ on its domain.
6. Prove $(\sin x + \cos x)^2 = 1 + \sin(2x)$.
7. Calculate $\cos(75^\circ)$ exactly by angle addition.
8. Express $\cos(2x)$ using $\sin x$ alone.
9. State the domain and range of $\arccos x$, and explain why the restriction $[0, \pi]$ is appropriate.
10. Evaluate $\arccos(\cos(5\pi/3))$ exactly.
11. Find $\tan(\arccos(4/5))$ exactly using a right triangle.
12. Prove $\sin(-x) = -\sin x$ geometrically and use it to compute $\sin(-\pi/3)$.
13. Verify the cosine difference identity by replacing β with $-\beta$ in the cosine sum identity.
14. Prove $\cos^2 x = (1 + \cos 2x)/2$ on all of $\mathbb{R}$.
15. Find $\sin 15^\circ$ exactly and state its sign.
16. Explain why $\arctan(\tan(3\pi/4))$ is not $3\pi/4$.

5 Trigonometric Equations and Applications

An inverse trig button gives an angle. An equation can ask for several angles, or for an entire repeating family. The difference comes from the geometry we have been studying: a rotating point can return to the same height or horizontal position more than once.

We will use that picture to solve equations, then use the same trigonometric relationships to find distances in triangles. Keep a sketch beside your algebra. It helps you decide how many answers to expect and whether your answers fit the situation.

5.1 Find the angles that give a chosen height

Consider $\sin x = \frac{1}{2}$. Sine is the vertical coordinate on the unit circle, so we want every point on the circle whose height is $\frac{1}{2}$. A horizontal line at that height meets the circle twice. The angles are $\pi/6$ and $5\pi/6$.

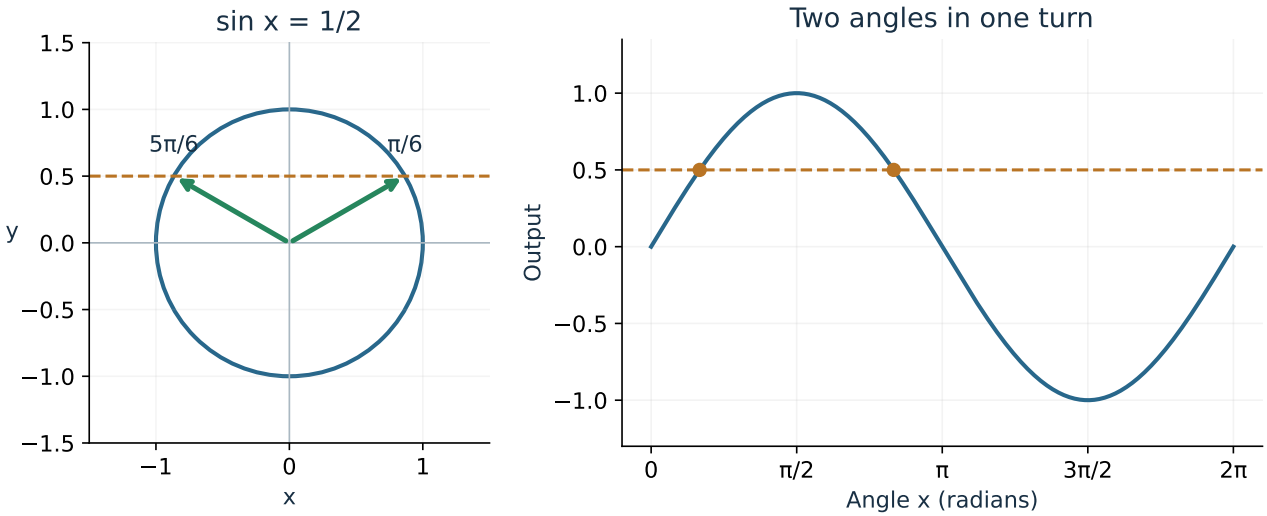


Figure 1: A horizontal line finds two points on the circle and two intersections on one sine cycle.

On the interval $0 \le x < 2\pi$, those are the two answers. The left endpoint is included; the right endpoint is excluded. That choice represents one complete turn without counting its starting position twice.

If the question asks for **all real solutions**, each angle may be followed by any whole number of turns:

$$x = \frac{\pi}{6} + 2\pi k \quad \text{or} \quad x = \frac{5\pi}{6} + 2\pi k, \qquad k \in \mathbb{Z}.$$

The integer k includes turns in either direction. For example, $k = -1$ gives negative angles that reach the same two points.

Worked example: a negative horizontal coordinate. Solve $\cos x = -\sqrt{2}/2$ on $[0, 2\pi)$. The reference angle is $\pi/4$. Cosine is negative on the left half of the circle, in Quadrants II and III. Therefore

$$x = \pi - \frac{\pi}{4} = \frac{3\pi}{4}, \qquad x = \pi + \frac{\pi}{4} = \frac{5\pi}{4}.$$

Both answers give the requested horizontal coordinate.

For tangent, a half-turn restores the same slope. Thus $\tan x = -1$ has the family $x = -\pi/4 + k\pi$. On $[0, 2\pi)$, the two members of that family are $3\pi/4$ and $7\pi/4$.

Try it. Solve $\sin x = -\sqrt{3}/2$ on $[0, 2\pi)$. Mark the two points before writing their angles.

5.2 Solve for the whole inside angle

In $\sin(2x) = \sqrt{3}/2$, the input to sine is $2x$. Call that input u. As x runs through $[0, 2\pi)$, the angle $u = 2x$ runs through $[0, 4\pi)$. We must search two turns.

On the first turn, $u = \pi/3$ or $2\pi/3$. Add 2π to get the second-turn answers $7\pi/3$ and $8\pi/3$. Now divide **each** by 2:

$$x = \frac{\pi}{6}, \quad \frac{\pi}{3}, \quad \frac{7\pi}{6}, \quad \frac{4\pi}{3}.$$

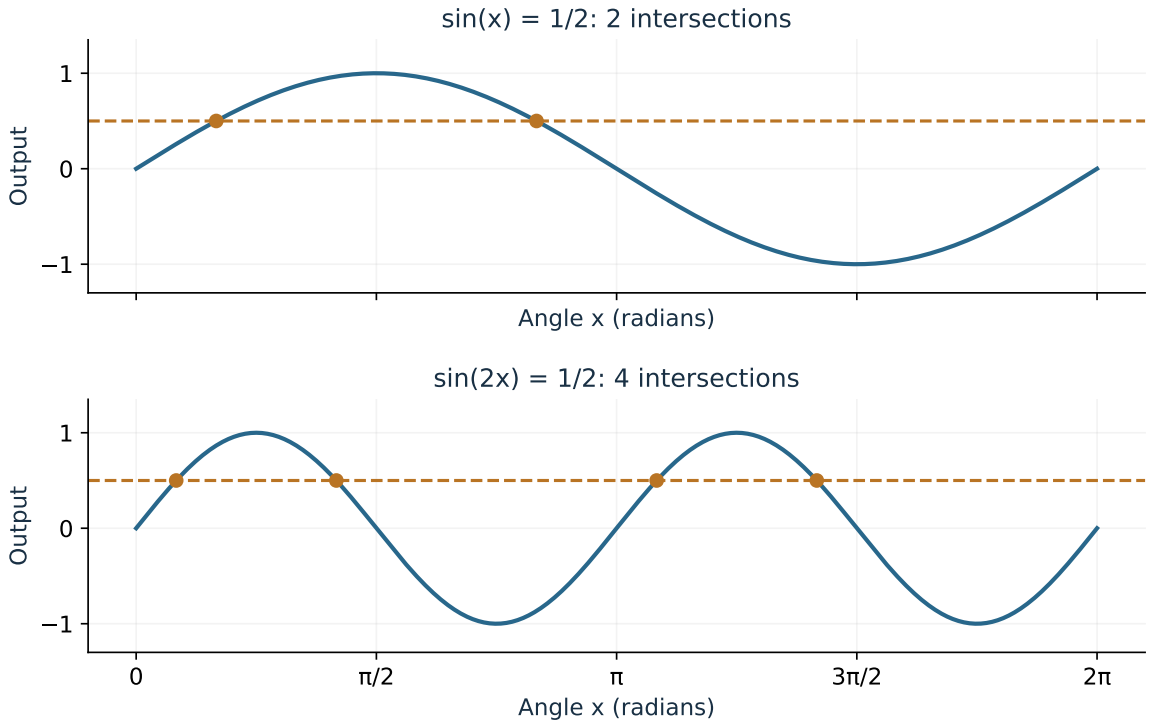


Figure 2: Doubling the inside angle fits two sine cycles into the same interval and doubles the intersections with height one half.

The graph provides a useful count: $\sin(2x)$ completes two cycles on this interval, and the chosen positive height occurs twice per cycle. We expect four solutions. At a maximum or minimum there is just one intersection per cycle; a target outside $[-1, 1]$ gives none.

Worked example: a shifted inside angle. Solve $\sin(x - \pi/4) = 1/2$ on $[0, 2\pi)$. Set $u = x - \pi/4$; its interval is $[-\pi/4, 7\pi/4)$. In that interval, $u = \pi/6$ or $5\pi/6$. Adding $\pi/4$ gives

$$x = \frac{5\pi}{12} \quad \text{or} \quad \frac{13\pi}{12}.$$

Substituting either answer restores one of the two familiar sine angles.

Use the Equation Explorer to compare frequencies 1, 2, and 3. Predict the number of intersections before displaying them.

5.3 Let algebra organize the choices

Start an equation such as $2\cos x - 1 = 0$ by isolating its trig expression. We obtain $\cos x = 1/2$, then use the circle to find $x = \pi/3, 5\pi/3$ on one turn.

Equations that are quadratic in a trig expression can be factored just as quadratics in an ordinary variable:

$$2\sin^2 x - \sin x = 0 \quad \Longrightarrow \quad \sin x(2\sin x - 1) = 0.$$

The zero-product rule gives two branches:

$$\sin x = 0 \quad \text{or} \quad \sin x = \frac{1}{2}.$$

The first branch contributes $0, \pi$; the second contributes $\pi/6, 5\pi/6$. There are four distinct solutions on $[0, 2\pi)$. Dividing the original equation by $\sin x$ would remove the first branch. Factoring keeps it visible.

An identity can also reveal a familiar equation. For $2\cos^2 x - 1 = 0$, we get $\cos x = \pm\sqrt{2}/2$. Each sign contributes two angles. When you use a quotient, reciprocal, or division, keep track of inputs excluded from the original expression. Check your final angles in that original expression.

Try it. Solve $2\cos^2 x - \cos x = 0$ on $[0, 2\pi)$. First write the two equations produced by factoring.

5.4 Label a triangle before choosing a law

For any triangle, use capital letters A, B, C for angles and the corresponding lowercase letters a, b, c for the sides **opposite** them. Angle A and side a are separated across the triangle. Angles add to $180°$, or π radians.

Right-triangle ratios use an opposite side, adjacent side, and hypotenuse. For a triangle with no right angle, we need relationships that work without a hypotenuse.

In the pictured triangle, the altitude has height $h = b\sin A = a\sin B$. Equating these expressions and rearranging gives $a/\sin A = b/\sin B$. The same reasoning with another altitude gives the **Law of Sines**:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}.$$

It is especially useful when an angle and its opposite side are both known. That complete pair supplies a ratio to compare with another pair.

Worked example: two angles and a side. Suppose $A = 40°$, $B = 65°$, and $a = 9$ cm. First find $C = 180° - 40° - 65° = 75°$. Pair a with A, and b with B:

$$\frac{b}{\sin 65°} = \frac{9}{\sin 40°}, \qquad b = \frac{9\sin 65°}{\sin 40°} \approx 12.69 \text{ cm}.$$

The larger angle $65°$ faces the longer side, which agrees with the result. Use degree mode for these measurements.

5.5 Why two triangles can fit the same measurements

Suppose $A = 30°$, $a = 5$, and $b = 8$. The known angle is opposite the side of length 5. The Law of Sines gives

$$\sin B = \frac{b\sin A}{a} = \frac{8\sin 30°}{5} = 0.8.$$

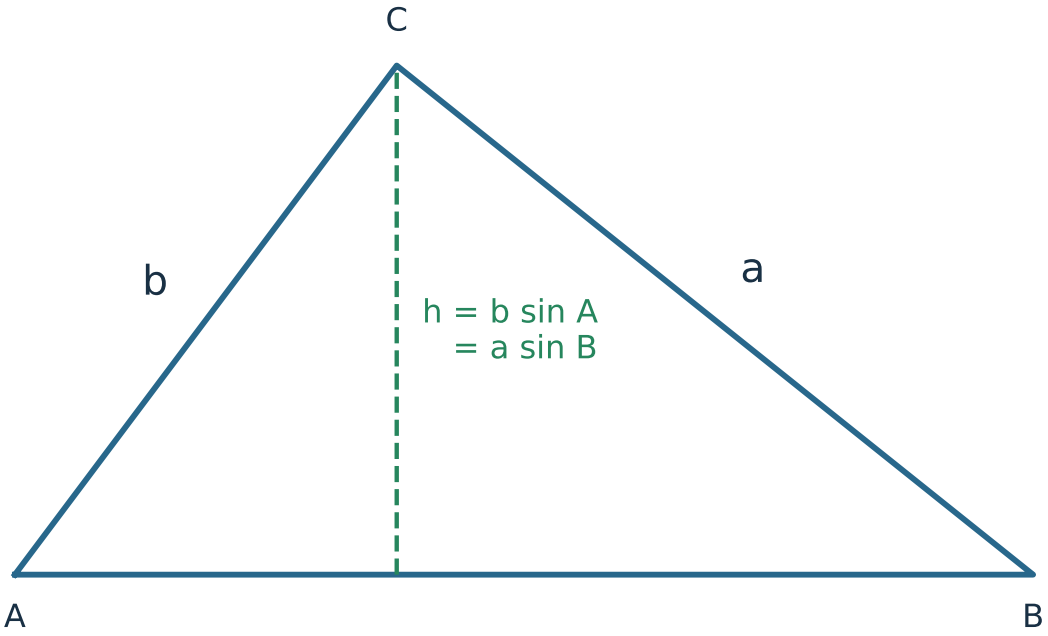


Figure 3: An altitude belongs to two right triangles. Expressing its height in two ways produces the Law of Sines.

The calculator gives $B_1 = \arcsin(0.8) \approx 53.13°$. Sine also has that value at its supplement, $B_2 = 180° - B_1 \approx 126.87°$. Check each candidate against the angle sum:

$$C_1 \approx 96.87°, \qquad C_2 \approx 23.13°.$$

Both remaining angles are positive, so both triangles work.

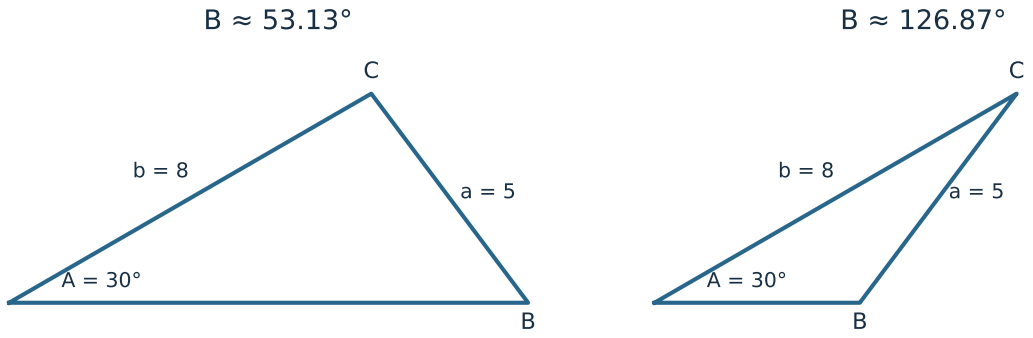


Figure 4: The same angle and two side lengths can form a shorter triangle or a longer triangle.

This is the **ambiguous SSA case**: two sides and a nonincluded angle are given. Follow three checks. If the calculated sine exceeds 1, no triangle exists. Otherwise find the principal angle and its supplement. Keep only distinct candidates for which $A + B < 180°$. Depending on the measurements, zero, one, or two triangles survive.

For example, changing a to 3 gives $\sin B = 4/3$, so there is no triangle. Changing a to 8 gives candidates 30° and 150°; the latter leaves $C = 0^\circ$, so only the first is a triangle.

5.6 Use the included angle with the Law of Cosines

If two sides and the angle **between** them are known, there is no complete opposite side-angle pair. The **Law of Cosines** supplies the missing side:

$$c^2 = a^2 + b^2 - 2ab\cos C.$$

To see where it comes from, place angle C at the origin. One endpoint is $(b, 0)$, and the other is $(a\cos C, a\sin C)$. Their squared distance is

$$\begin{aligned} c^2 &= (a\cos C - b)^2 + (a\sin C)^2 \\ &= a^2(\cos^2 C + \sin^2 C) + b^2 - 2ab\cos C. \end{aligned}$$

The Pythagorean identity reduces the first term to a^2.

Worked example: a connector between two roads. Two roads extend 7 km and 10 km from a junction, separated by 60°. The connector between their endpoints has length

$$c = \sqrt{7^2 + 10^2 - 2(7)(10)\cos 60^\circ} = \sqrt{79} \approx 8.89 \text{ km}.$$

At 90°, the cosine term is zero and the formula becomes the Pythagorean theorem. For the same side lengths, a wider included angle produces a longer connector.

If all three sides are known, rearrange the law to recover an angle:

$$C = \arccos\left(\frac{a^2 + b^2 - c^2}{2ab}\right).$$

The geometric angle lies between 0° and 180°, which matches the range of arccosine.

5.7 Choose a method and interpret the result

For an equation, decide which interval to search, solve for the trig expression, and use periodicity to find every answer. For a triangle, draw and label the known measurements. A complete opposite pair suggests the Law of Sines; two sides with their included angle, or three sides, suggest the Law of Cosines.

A shoreline survey illustrates the choice. Two observers stand 200 m apart and measure interior sight angles of 47° and 68° to an island marker. The third angle is 65°. The known baseline is opposite that angle, so it supplies the complete pair for the Law of Sines. The geometry tells us which measurements belong together before we calculate.

5.8 Review exercises

1. Solve $\sin x = -\sqrt{3}/2$ on $[0, 2\pi)$, then write all real solutions.
2. Solve $2\sin(3x) = 1$ on $[0, 2\pi)$. Explain the number of answers.
3. Solve $2\cos^2 x - \cos x = 0$ on $[0, 2\pi)$.
4. Solve $\tan x = -1$ for all real x.
5. Solve $\cos(2x) = 1/2$ on $[0, 2\pi)$.
6. A triangle has $A = 35°$, $B = 80°$, $a = 12$. Find C and b.
7. For $A = 30°$, $a = 5$, $b = 8$, find both possible values of C.
8. Find the third side for sides 6 and 9 with included angle $120°$.
9. Explain how the Law of Cosines includes the Pythagorean theorem.
10. Solve $\sin(x - \pi/6) = 1/2$ on $[0, 2\pi)$.

5.9 Check your understanding

The first Try it has answers $4\pi/3, 5\pi/3$. For the factoring Try it, $\cos x = 0$ or $\cos x = 1/2$, giving $\pi/2, 3\pi/2, \pi/3, 5\pi/3$. Each branch contributes its own angles. A sketch or a substitution checks the result.

6 Vectors and Geometric Applications

A person walks 6 km east, then 8 km north. The route is 14 km long, but the endpoint is only 10 km from the start. To describe where the person ends up, we need both a distance and a direction. A vector records those two pieces of information together.

This chapter develops vectors from familiar coordinates and triangle geometry. We will add movements, resolve forces into components, and measure how much one vector points along another.

6.1 Describe a displacement with components

The displacement $\mathbf{u} = \langle 6, 8\rangle$ means 6 units horizontally and 8 units vertically. Angle brackets distinguish a vector from a location $(6, 8)$. The same components can describe an arrow starting anywhere in the plane.

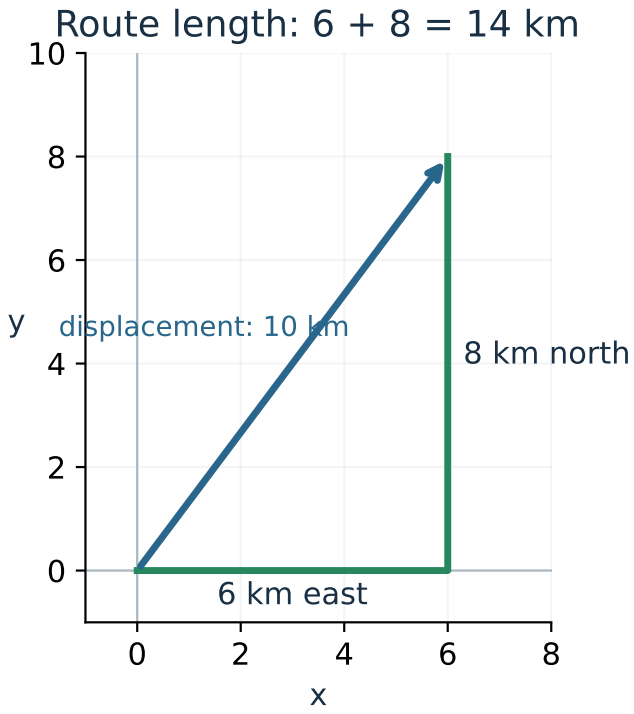


Figure 5: The walking route is 14 km; the displacement joins the start directly to the endpoint.

For a vector $\mathbf{u} = \langle u_x, u_y\rangle$, its **magnitude** is the length of its arrow:

$$\|\mathbf{u}\| = \sqrt{u_x^2 + u_y^2}.$$

For the walk, $\sqrt{6^2 + 8^2} = 10$ km. This is displacement magnitude. The distance walked depends on the route.

To find the vector from $P = (p_x, p_y)$ to $Q = (q_x, q_y)$, subtract start from finish:

$$\overrightarrow{PQ} = \langle q_x - p_x, q_y - p_y\rangle.$$

Starting at P, adding those two changes lands exactly at Q.

Worked example: from one location to another. For $P = (-2, 3)$, $Q = (5, -1)$,

$$\overrightarrow{PQ} = \langle 5 - (-2), -1 - 3\rangle = \langle 7, -4\rangle, \qquad \|\overrightarrow{PQ}\| = \sqrt{65}.$$

The endpoint lies 7 units right and 4 units down from the start. Reversing the journey gives $\overrightarrow{QP} = \langle -7, 4 \rangle$.

6.2 Add arrows by following one with another

If a movement $\mathbf{u}$ is followed by $\mathbf{v}$, the total horizontal change is $u_x + v_x$, and the total vertical change is $u_y + v_y$:

$$\mathbf{u} + \mathbf{v} = \langle u_x + v_x, u_y + v_y \rangle.$$

Draw the second arrow with its tail at the first arrow's head. The sum joins the original start to the final endpoint. Translating an arrow for this construction preserves its components.

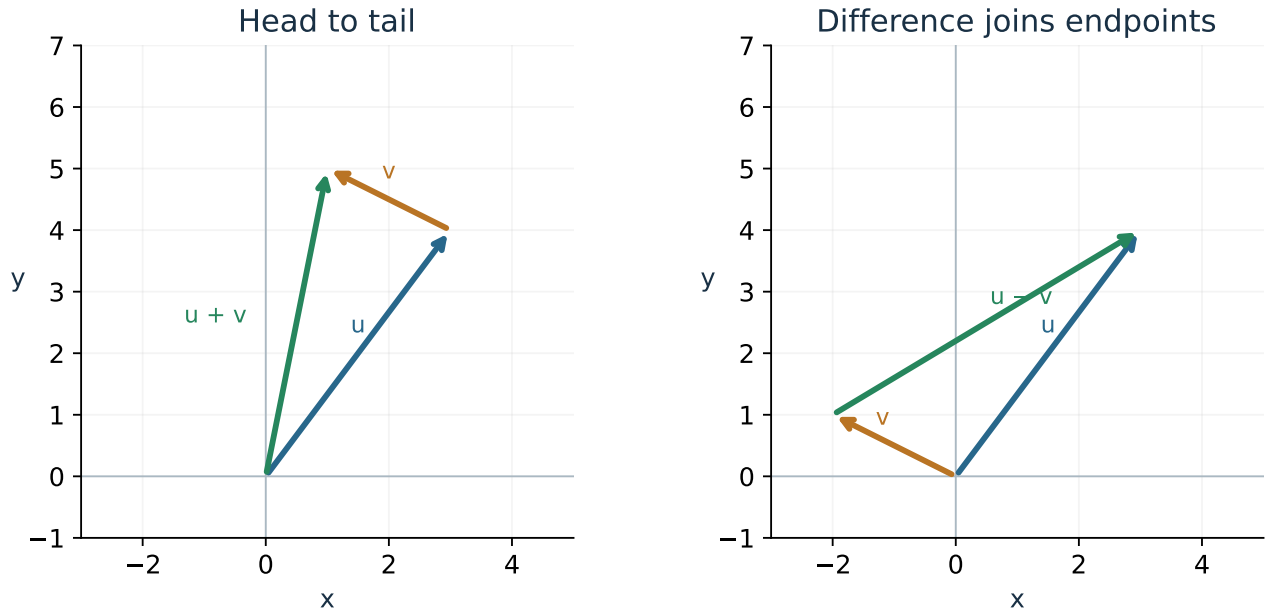


Figure 6: Head-to-tail addition and the parallelogram give the same sum. The difference joins the endpoints when both arrows start together.

For $\mathbf{u} = \langle 3, 4 \rangle$ and $\mathbf{v} = \langle -2, 1 \rangle$, the sum is $\langle 1, 5 \rangle$. The second movement goes left by 2 and up by 1. Adding it reduces the eastward component and increases the northward component.

Multiplying by a number scales every component:

$$c\mathbf{u} = \langle cu_x, cu_y \rangle.$$

For $c > 0$, the arrow keeps its direction; for $c < 0$, it reverses direction. Its length becomes $|c|\|\mathbf{u}\|$. The zero vector $\langle 0, 0 \rangle$ has magnitude zero and has no direction.

Subtraction means adding the opposite: $\mathbf{u} - \mathbf{v} = \mathbf{u} + (-\mathbf{v})$. If both vectors start at the origin, $\mathbf{u} - \mathbf{v}$ points from the endpoint of $\mathbf{v}$ to the endpoint of $\mathbf{u}$.

Try it. With $\mathbf{u} = \langle 2, 5 \rangle$, $\mathbf{v} = \langle -1, 3 \rangle$, calculate $\mathbf{u} + \mathbf{v}$ and $3\mathbf{u} - 2\mathbf{v}$. Check the direction of each component.

6.3 Convert between length-and-angle and components

A vector of magnitude $r \geq 0$ at angle θ, measured counterclockwise from the positive horizontal axis, has components

$$\mathbf{u} = \langle r \cos\theta, r \sin\theta \rangle.$$

This is the unit-circle point scaled by r. The magnitude returns to r, because $\cos^2\theta + \sin^2\theta = 1$.

Worked example: a force pointing up and left. A 20 N force acts at 150°:

$$\mathbf{F} = \langle 20\cos 150^\circ, 20\sin 150^\circ\rangle = \langle -10\sqrt{3}, 10\rangle \text{ N}.$$

The negative horizontal component describes the leftward part of the force; the positive vertical component describes its upward part.

For a vector with known components, find the magnitude first. To recover direction, inspect the quadrant before using inverse tangent. For $\langle -3, 4\rangle$, the reference angle is $\arctan(4/3) \approx 53.13^\circ$. The vector lies in Quadrant II, so its direction is $180^\circ - 53.13^\circ = 126.87^\circ$. The single quotient $4/(-3)$ does not by itself specify that quadrant.

A **unit vector** has magnitude 1. For a nonzero vector,

$$\widehat{\mathbf{u}} = \frac{\mathbf{u}}{\|\mathbf{u}\|}.$$

Thus $\langle -3, 4\rangle/5 = \langle -3/5, 4/5\rangle$ keeps the direction and sets the length to 1.

6.4 The dot product measures alignment

The dot product combines two vectors into a number:

$$\mathbf{u}\cdot\mathbf{v} = u_x v_x + u_y v_y.$$

For nonzero vectors, this number also has a geometric form:

$$\mathbf{u}\cdot\mathbf{v} = \|\mathbf{u}\|\|\mathbf{v}\|\cos\phi,$$

where $0 \le \phi \le \pi$ is the angle between their arrows.

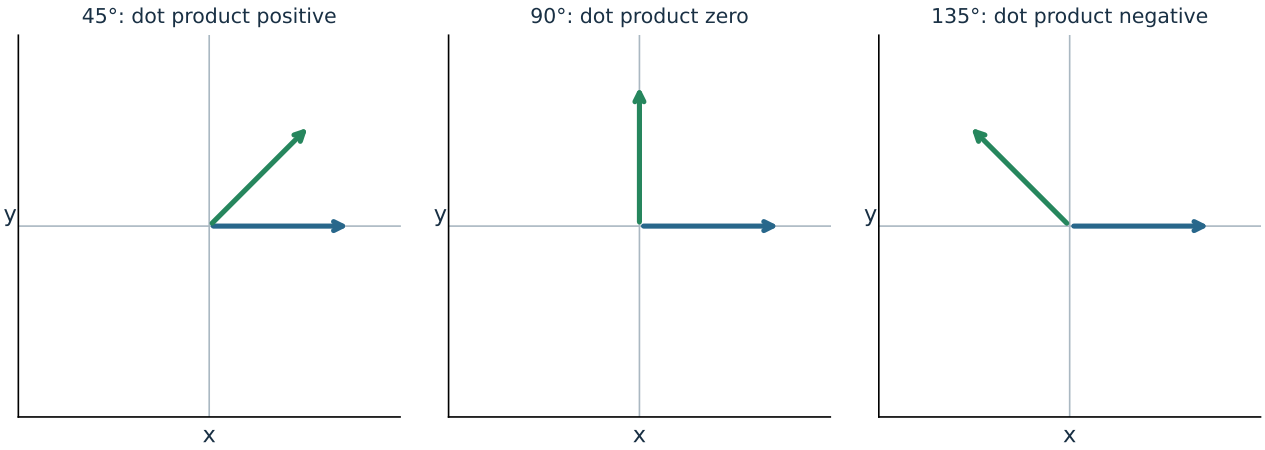


Figure 7: For vectors of fixed lengths, the dot product is positive at an acute angle, zero at a right angle, and negative at an obtuse angle.

The sign has meaning. Positive indicates an acute angle, zero indicates a right angle, and negative indicates an obtuse angle. A zero vector has dot product zero with every vector, but there is no angle to assign to it.

Worked example: perpendicular vectors. Let $\mathbf{u} = \langle 3, 4\rangle$, $\mathbf{v} = \langle 4, -3\rangle$. Their dot product is $12 - 12 = 0$. Both are nonzero, so the angle between them is 90°.

To calculate another angle, solve the geometric formula:

$$\phi = \arccos\left(\frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{u}\|\|\mathbf{v}\|}\right).$$

For $\langle 2,1\rangle$ and $\langle 1,2\rangle$, the dot product is 4 and each magnitude is $\sqrt{5}$. Thus $\phi = \arccos(4/5) \approx 36.87°$.

6.5 A projection is the part pointing along a chosen direction

Imagine a light casting an arrow's shadow onto a line through the origin. The **projection** records the part of $\mathbf{u}$ along the direction of a nonzero vector $\mathbf{v}$:

$$\operatorname{proj}_{\mathbf{v}}\mathbf{u} = \frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{v}\|^2}\mathbf{v}.$$

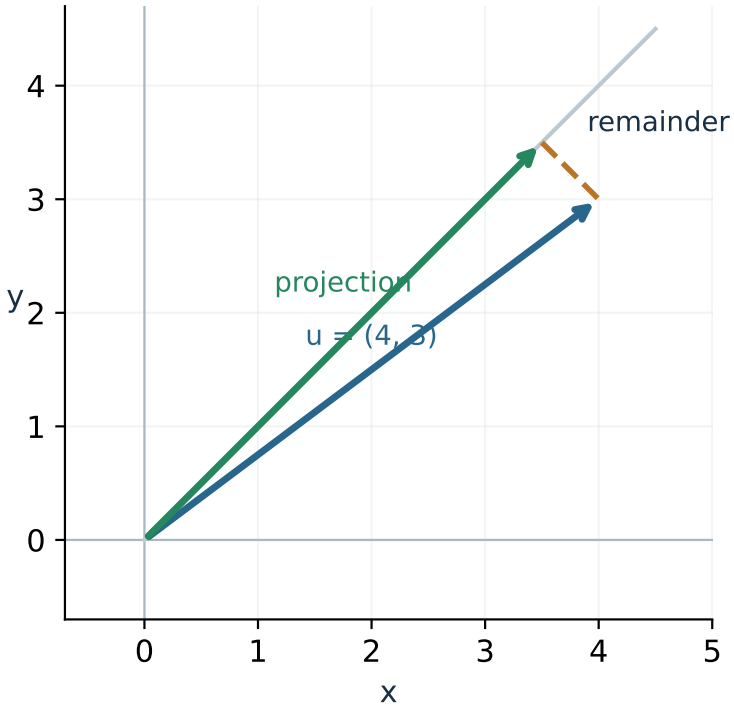


Figure 8: Projection separates a vector into a part parallel to the chosen direction and a perpendicular remainder.

Why this coefficient? First divide $\mathbf{v}$ by its length to get a unit direction $\hat{\mathbf{v}}$. The signed length of the shadow is $\mathbf{u}\cdot\hat{\mathbf{v}} = \|\mathbf{u}\|\cos\phi$. Multiply that length by $\hat{\mathbf{v}}$, obtaining the displayed vector formula.

Worked example: horizontal and slanted projections. Projecting $\langle 4,3\rangle$ onto $\langle 1,0\rangle$ gives $\langle 4,0\rangle$. Onto $\mathbf{v} = \langle 1,1\rangle$, the coefficient is $(4+3)/(1+1) = 7/2$, giving $\langle 7/2, 7/2\rangle$. The remainder is $\langle 1/2, -1/2\rangle$; its dot product with $\langle 1,1\rangle$ is zero.

6.6 Work uses the force along the displacement

For a constant force $\mathbf{F}$ acting through displacement $\mathbf{d}$, the work is

$$W = \mathbf{F}\cdot\mathbf{d} = \|\mathbf{F}\|\|\mathbf{d}\|\cos\phi.$$

Newtons times meters give joules. A perpendicular force contributes zero work to that displacement. A force pointing against the displacement contributes negative work.

A force $\langle 12, 5\rangle$ N acting through $\langle 3, 0\rangle$ m does $12(3) + 5(0) = 36$ J of work. Only the horizontal component contributes. If the force instead acted at 60° to the displacement, only half its magnitude would act along the movement.

Try it. A 50 N force acts at 60° to a 4 m displacement. Calculate its work and compare with the work from a parallel force of the same magnitude.

6.7 The triangle law returns in vector language

Place two vectors $\mathbf{u}, \mathbf{v}$ at a common starting point. Their difference joins their endpoints. Expanding its squared magnitude gives

$$\begin{aligned}\|\mathbf{u} - \mathbf{v}\|^2 &= (\mathbf{u} - \mathbf{v}) \cdot (\mathbf{u} - \mathbf{v}) \\ &= \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - 2\mathbf{u} \cdot \mathbf{v} \\ &= \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - 2\|\mathbf{u}\|\|\mathbf{v}\| \cos\phi.\end{aligned}$$

This is the Law of Cosines. The sides are the two original vectors and their connecting difference. It explains why component calculations and triangle calculations agree.

Use the Vector Explorer to compare the picture with the component arithmetic. When you change one component, describe the movement of the endpoint before reading the new magnitude or dot product.

6.8 Review exercises

1. Find the vector from $(-2, 3)$ to $(5, -1)$, and its magnitude.
2. For $\mathbf{u} = \langle 2, 5\rangle$, $\mathbf{v} = \langle -1, 3\rangle$, compute $\mathbf{u} + \mathbf{v}$ and $3\mathbf{u} - 2\mathbf{v}$.
3. Find exact components of a displacement of magnitude 12 at 150°.
4. Find a unit vector in the direction $\langle -3, 4\rangle$.
5. Compute $\langle 2, -1\rangle \cdot \langle 5, 3\rangle$ and interpret its sign.
6. Determine whether $\langle 6, 2\rangle$ and $\langle -1, 3\rangle$ are perpendicular.
7. Find the projection of $\langle 4, 3\rangle$ onto $\langle 1, 0\rangle$.
8. Find the work of a 50 N force acting at 60° to a 4 m displacement.
9. Find the angle between $\langle 1, 0\rangle$ and $\langle 1, \sqrt{3}\rangle$.
10. Give two routes with the same displacement but different total distances.

6.9 Check your understanding

The first Try it gives $\langle 1, 8\rangle$ and $\langle 8, 9\rangle$. The work example gives $50(4)\cos 60^\circ = 100$ J, compared with 200 J for a parallel force. The factor $\cos 60^\circ = 1/2$ measures the contributing part of the force.

7 Polar Coordinates and Complex Numbers

An address such as $(3, 4)$ describes a point by moving horizontally and vertically. A distance of 5 at a direction of about $53°$ describes the same point by aiming and traveling. Polar coordinates use that second description.

Complex numbers will add another interpretation. They represent points in the plane, and their multiplication performs rotations and scaling. The trigonometry from earlier chapters explains why.

7.1 Locate a point by distance and direction

A polar pair (r, θ) gives a radius and an angle from the positive horizontal axis. For $r > 0$, start along the ray at angle θ and travel r units from the origin. The familiar circle coordinates give

$$x = r\cos\theta, \qquad y = r\sin\theta.$$

These are the same component formulas used for vectors. Here they locate a point.

Worked example: a point in Quadrant II. For $(r, \theta) = (4, 5\pi/6)$,

$$x = 4(-\sqrt{3}/2) = -2\sqrt{3}, \qquad y = 4(1/2) = 2.$$

The negative x and positive y agree with the angle's quadrant. The point is 4 units from the origin.

To go in the other direction with $r \geq 0$, use $r = \sqrt{x^2 + y^2}$. Find the angle from both coordinate signs. For $(-3, -3)$, the distance is $3\sqrt{2}$, and the direction is $5\pi/4$ on $[0, 2\pi)$. Computing $\arctan(y/x) = \pi/4$ alone would place the point in the wrong quadrant.

Try it. Convert $(6, -\pi/3)$ to Cartesian coordinates. Give the same direction using an angle between 0 and 2π.

7.2 Several polar addresses can name one point

Adding a full turn changes an angle's description while leaving its direction fixed:

$$(r, \theta) \quad \text{and} \quad (r, \theta + 2\pi k)$$

represent the same point for every integer k.

A negative radius means travel in the direction opposite the chosen ray. Consequently,

$$(r, \theta) \quad \text{and} \quad (-r, \theta + \pi)$$

also represent the same point.

For example, $(2, \pi/3)$, $(2, 7\pi/3)$, and $(-2, 4\pi/3)$ all locate $(1, \sqrt{3})$. Check the last address directly: both cosine and sine at $4\pi/3$ are negative, and multiplying by -2 makes the coordinates positive.

At $r = 0$, all choices of angle locate the origin. This redundancy is part of the coordinate system. A unique polar address requires conventions about radius and angle.

7.3 Put complex numbers on the plane

A complex number $z = a + bi$, where $i^2 = -1$, corresponds to the point (a, b). The horizontal axis records its real part a, and the vertical axis its imaginary part b.

Its **modulus** is its distance from zero:

$$|z| = \sqrt{a^2 + b^2}.$$

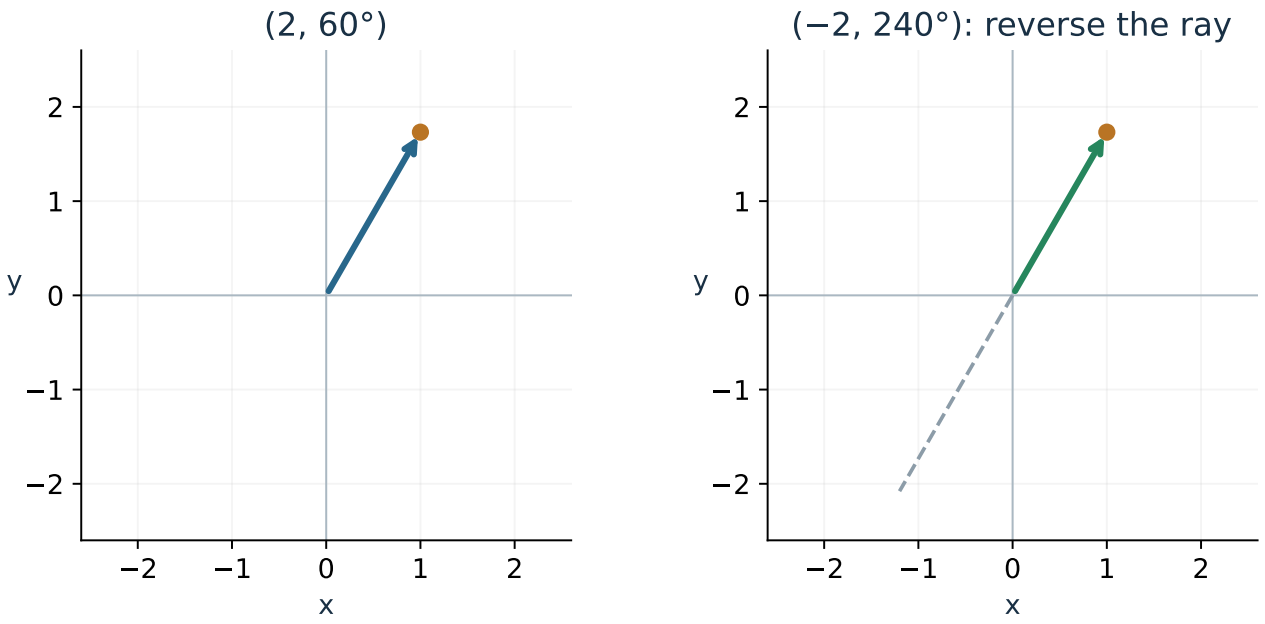


Figure 9: A positive radius on one ray and a negative radius on the opposite ray reach the same location.

For $z \neq 0$, an **argument** is an angle locating its direction. Polar form writes

$$z = r(\cos\theta + i\sin\theta), \qquad r = |z|.$$

Arguments differ by whole turns. The number zero has modulus zero and no defined argument.

Worked example: rectangular to polar form. For $z = -\sqrt{3} + i$, the modulus is $\sqrt{3+1} = 2$. The point lies in Quadrant II with reference angle $\pi/6$, so one argument is $5\pi/6$:

$$z = 2\left(\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6}\right).$$

Multiplying out the two components returns the original number.

7.4 Addition combines components; multiplication combines transformations

Complex addition follows vector addition:

$$(a + bi) + (c + di) = (a + c) + (b + d)i.$$

For multiplication, expand and use $i^2 = -1$:

$$(a + bi)(c + di) = (ac - bd) + (ad + bc)i.$$

For example, $(1 + i)(1 - i) = 1 - i^2 = 2$, while $(1 + i) + (1 - i) = 2$ as well. These particular results agree, but the two operations follow different rules.

To see the geometric rule for multiplication, write

$$z = r(\cos\alpha + i\sin\alpha), \qquad w = s(\cos\beta + i\sin\beta).$$

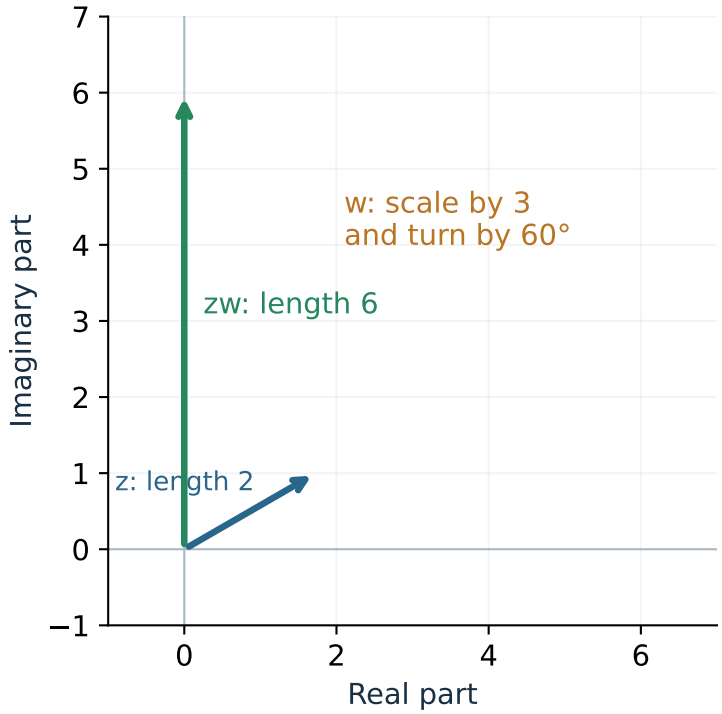


Figure 10: Multiplication by a complex factor rotates the original vector and scales its length.

Expanding gives real part $rs(\cos\alpha\cos\beta - \sin\alpha\sin\beta)$ and imaginary part $rs(\sin\alpha\cos\beta + \cos\alpha\sin\beta)$. The angle-addition identities turn these into

$$zw = rs\big(\cos(\alpha+\beta) + i\sin(\alpha+\beta)\big).$$

Multiply the moduli and add the arguments.

Worked example: rotate and stretch. Suppose z has modulus 2 and argument 30°, while w has modulus 3 and argument 60°. Then zw has modulus 6 and argument 90°, so $zw = 6i$.

Multiplication by i gives a quarter-turn counterclockwise: $i(a+bi) = -b + ai$, so (a, b) becomes $(-b, a)$. Its length stays fixed. Multiplication by -1 gives a half-turn. Multiplication by a positive real number changes length while keeping direction.

Use the Polar and Complex Explorer to predict a product's length and direction before displaying its rectangular coordinates.

7.5 Powers repeat the same rotation and scaling

Raising a number to a positive integer power repeats its multiplication rule:

$$\big[r(\cos\theta + i\sin\theta)\big]^n = r^n(\cos n\theta + i\sin n\theta).$$

This is **De Moivre's theorem**. It also holds for zero and negative integer powers when the original number is nonzero.

For example, $(\cos(\pi/6) + i\sin(\pi/6))^3$ has modulus 1 and angle $\pi/2$, so it equals i. The exponent multiplies the original angle by 3.

To find roots, reverse those two changes. Suppose

$$z^n = R(\cos\phi + i\sin\phi), \qquad R > 0.$$

The root modulus is $R^{1/n}$. For the angles, include all complete turns before dividing:

$$\theta_k = \frac{\phi + 2\pi k}{n}, \qquad k = 0, 1, \ldots, n-1.$$

The roots are equally spaced around a circle.

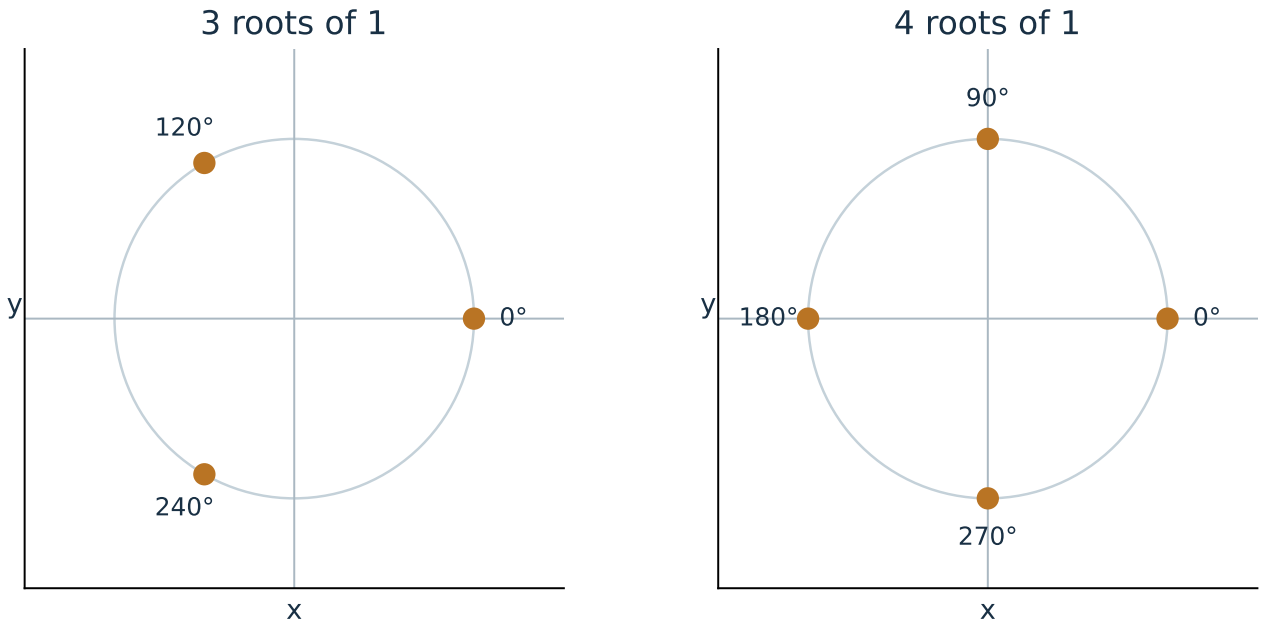


Figure 11: The cube roots of one are separated by 120 degrees; the fourth roots are separated by 90 degrees.

Worked example: three cube roots of 1. Use $R = 1, \phi = 0, n = 3$. The angles are $0, 2\pi/3, 4\pi/3$, giving

$$1, \qquad -\frac{1}{2} + \frac{\sqrt{3}}{2}i, \qquad -\frac{1}{2} - \frac{\sqrt{3}}{2}i.$$

Cubing any of these multiplies its angle by 3, producing a whole number of turns.

For the four fourth roots of 1, the angles are $0, \pi/2, \pi, 3\pi/2$, giving $1, i, -1, -i$.

Try it. Find all square roots of $4i$. Begin with modulus 4 and argument $\pi/2$.

7.6 A polar equation draws points by changing the radius

In a polar equation such as $r = 2\cos\theta$, each angle supplies a radius. Compute a few pairs, convert them to points, and look for the curve. When the formula supplies a negative radius, reverse the ray as described earlier.

Algebra identifies this particular curve. Multiply by r, then use $r^2 = x^2 + y^2$ and $r\cos\theta = x$:

$$r = 2\cos\theta \implies x^2 + y^2 = 2x \implies (x-1)^2 + y^2 = 1.$$

It is a circle with center $(1, 0)$ and radius 1. Multiplying by r deserves a check at $r = 0$; here the origin belongs to the original curve, since $\cos\theta = 0$ for suitable angles.

The curve is traced once as θ runs from 0 to π. From $\pi/2$ onward, the radius is negative, but the resulting points continue around the lower half of the same circle.

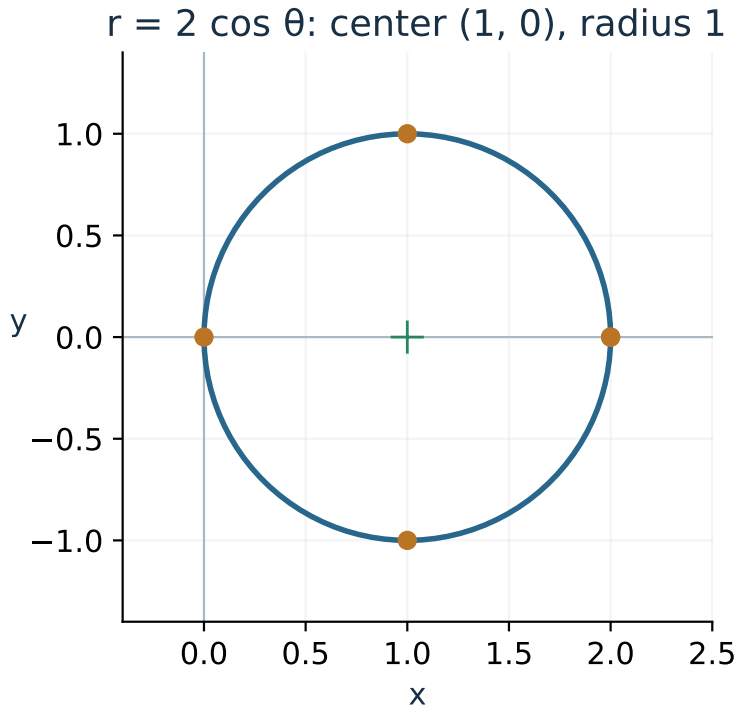


Figure 12: Sample polar coordinates for r equal to twice cosine trace a circle whose center is displaced from the origin.

7.7 Review exercises

1. Convert $(5, 2\pi/3)$ to Cartesian coordinates.
2. Give two distinct polar representations of $(\sqrt{2}, \sqrt{2})$, including one with negative radius.
3. Write $-1 - \sqrt{3}i$ in polar form with argument in $[0, 2\pi)$.
4. Calculate $(2 + i)(1 - 3i)$ in rectangular form.
5. Describe multiplication by $-i$ geometrically.
6. Evaluate $(\cos(\pi/4) + i\sin(\pi/4))^4$.
7. Find and plot all three cube roots of 1.
8. Convert $r = 4\sin\theta$ to Cartesian form and identify the curve.
9. Find all square roots of -9.
10. Explain what happens to modulus and argument when a nonzero complex number is squared.

7.8 Check your understanding

The first Try it gives $(3, -3\sqrt{3})$, with positive angle $5\pi/3$. The square roots of $4i$ have modulus 2 and arguments $\pi/4, 5\pi/4$; in rectangular form they are $\sqrt{2} + \sqrt{2}i$ and $-\sqrt{2} - \sqrt{2}i$.

8 Parametric Equations and Conic Sections

A curve shows where a moving object can go. To describe its motion, we also need to know where it is at each time. Parametric equations supply that information by letting both coordinates depend on one parameter.

We will use moving points to understand curves, then examine the distance rules defining circles, ellipses, parabolas, and hyperbolas. These are different ways to describe the same geometric objects.

8.1 Let two coordinates depend on one input

A parametric curve has the form

$$x = f(t), \qquad y = g(t).$$

Choose a value of t, calculate both coordinates, and plot their ordered pair. If t represents time, that pair is a position at that instant.

For $x = t + 2,\ y = 3t - 1$, the values $t = -1, 0, 1$ give $(1, -4), (2, -1), (3, 2)$. The point moves right and up as t increases. Solving $t = x - 2$ and substituting gives $y = 3x - 7$, so the path is a line.

Eliminating the parameter means finding an equation involving only x and y. Keep the original interval for t in mind. For $0 \le t \le 1$, the preceding equations trace only the segment from $(2, -1)$ to $(3, 2)$.

8.2 The same curve can carry different motions

For $x = \cos t,\ y = \sin t$, the point moves counterclockwise around the unit circle. At $t = 0$ it is $(1, 0)$; at $t = \pi/2$ it reaches $(0, 1)$.

Replace t by $2t$. The point still lies on the unit circle, because $\cos^2(2t) + \sin^2(2t) = 1$. It now completes a revolution in π units of time.

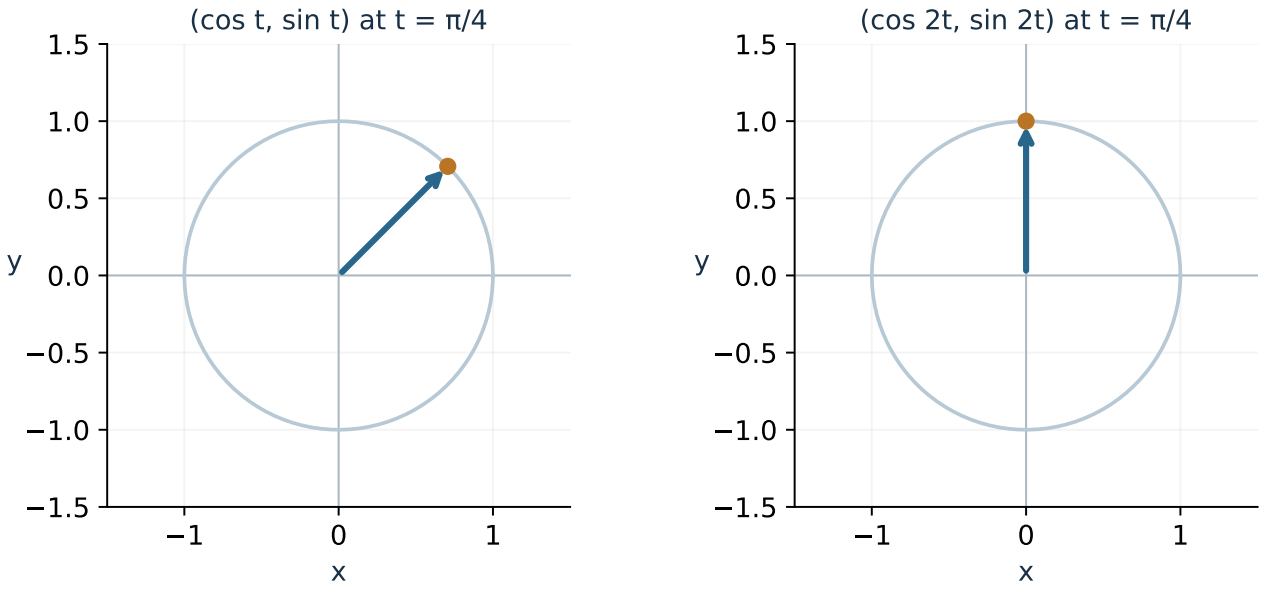


Figure 13: The two motions occupy the same circle but reach different positions at the same time.

On $0 \le t \le 2\pi$, $(\cos t, \sin t)$ travels around once, while $(\cos 2t, \sin 2t)$ travels twice. Replacing t by $-t$ reverses the direction. The Cartesian equation $x^2 + y^2 = 1$ describes all three paths, but contains no timing information.

Worked example: eliminate a trig parameter. Suppose

$$x = 4\cos t, \qquad y = 3\sin t.$$

Divide by the two scale factors, square, and add:

$$\left(\frac{x}{4}\right)^2 + \left(\frac{y}{3}\right)^2 = \cos^2 t + \sin^2 t = 1.$$

Thus the curve is $x^2/16 + y^2/9 = 1$. A full turn of t traces the whole ellipse; $0 \le t \le \pi$ traces its upper half.

Try it. Eliminate t from $x = 2t + 1,\ y = t - 3$, with $0 \le t \le 4$. Give both the line equation and the endpoints of the traced segment.

8.3 Follow a projectile with two equations

In an idealized projectile model, horizontal velocity stays constant while gravity changes the vertical motion. If the object begins at (x_0, y_0), with launch speed v_0 at angle θ, then

$$x(t) = x_0 + v_0\cos\theta\, t, \qquad y(t) = y_0 + v_0\sin\theta\, t - \frac{1}{2}gt^2.$$

Use $g = 9.8$ m/s^2 when distances are in meters and time is in seconds. The model ignores air resistance.

Worked example: a launch from the origin. Let $v_0 = 20$ m/s and $\theta = 30°$. Then

$$x(t) = 10\sqrt{3}\, t, \qquad y(t) = 10t - 4.9t^2.$$

At $t = 1$, the position is $(10\sqrt{3}, 5.1)$ meters. To find the landing time on level ground, set $y(t) = 0$:

$$t(10 - 4.9t) = 0.$$

The launch is at $t = 0$; landing occurs at $t = 100/49 \approx 2.04$ s. The physical trajectory uses that time interval.

Eliminating $t = x/(10\sqrt{3})$ produces a quadratic equation in x, so the path is a parabola. The time equations additionally tell us which direction the object moves and when it reaches each point.

8.4 A circle fixes one distance

A circle is the set of points a fixed distance $r > 0$ from a center (h, k). The distance formula gives

$$(x - h)^2 + (y - k)^2 = r^2.$$

For $(x - 2)^2 + (y + 1)^2 = 16$, the center is $(2, -1)$ and the radius is 4. A parameterization is $x = 2 + 4\cos t,\ y = -1 + 4\sin t$.

The signs inside the squares tell you which point makes both squared terms zero. This is a reliable way to read the center.

8.5 An ellipse fixes a sum of distances

An ellipse consists of points whose distances to two fixed **foci** add to a constant. Imagine a loop of string around two pins: holding the string taut with a pencil traces an ellipse.

For an ellipse centered at (h, k) with horizontal major axis,

$$\frac{(x - h)^2}{a^2} + \frac{(y - k)^2}{b^2} = 1, \qquad a > b > 0.$$

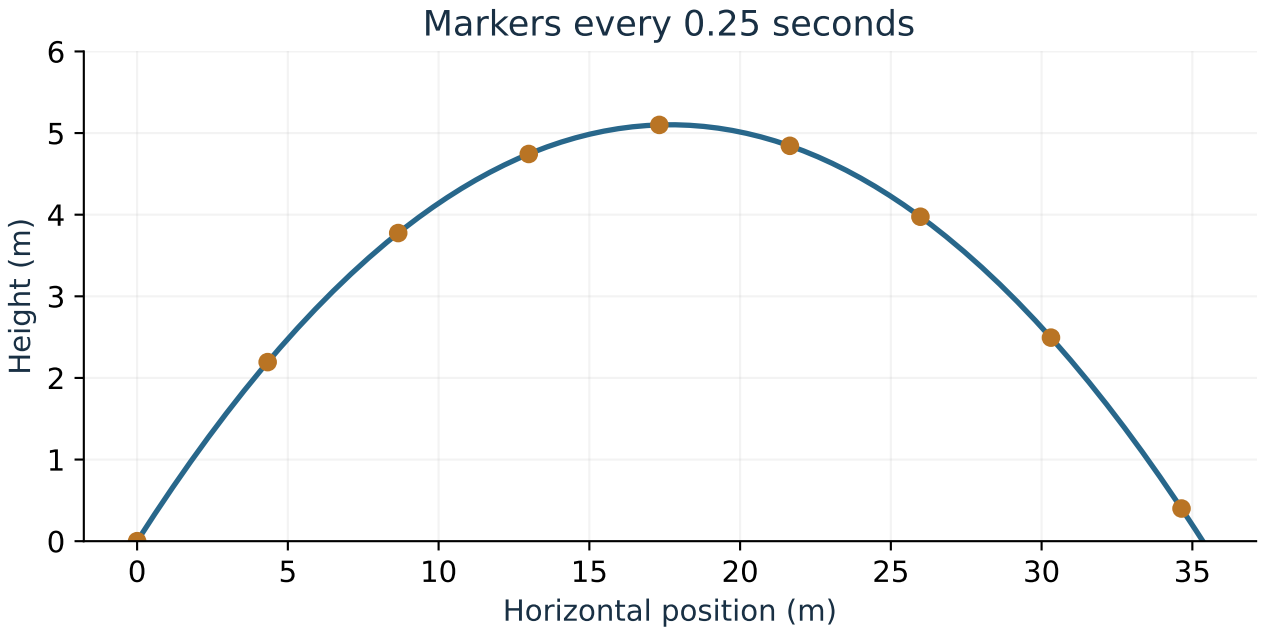


Figure 14: Equal time steps produce equal horizontal changes, while vertical changes vary under gravity.

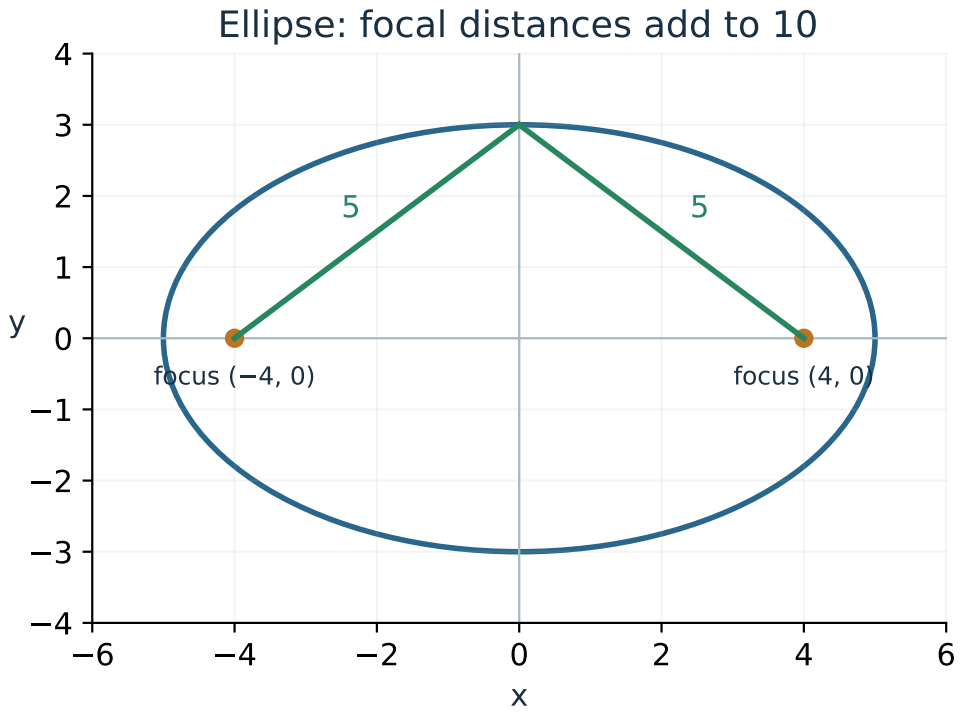


Figure 15: The two focal distances add to 2a at every point of the ellipse. At the top, the distance formula gives a squared equal to b squared plus c squared.

Its horizontal semiaxis has length a, and its vertical semiaxis length b. The foci are $(h-c,k),(h+c,k)$, with

$$c^2 = a^2 - b^2.$$

At a horizontal vertex, the two focal distances add to $2a$. At the top of the ellipse, symmetry makes each distance a; the right triangle there gives $a^2 = b^2 + c^2$.

Worked example: identify the ellipse. For $x^2/25 + y^2/9 = 1$, the semiaxes are 5 and 3. Thus $c = \sqrt{25-9} = 4$, and the foci are $(\pm 4, 0)$. Every point on the curve has total focal distance 10.

If the larger denominator is under the y-term, the major axis is vertical. Keep a for the larger semiaxis; then the foci move vertically from the center. Equal denominators give a circle, with coincident foci.

8.6 A parabola balances distance to a point and a line

A parabola consists of points equally far from a focus and a line called the **directrix**. For

$$(x-h)^2 = 4p(y-k), \qquad p \neq 0,$$

the vertex is (h,k), focus $(h,k+p)$, and directrix $y = k-p$. The sign of p determines whether the curve opens upward or downward.

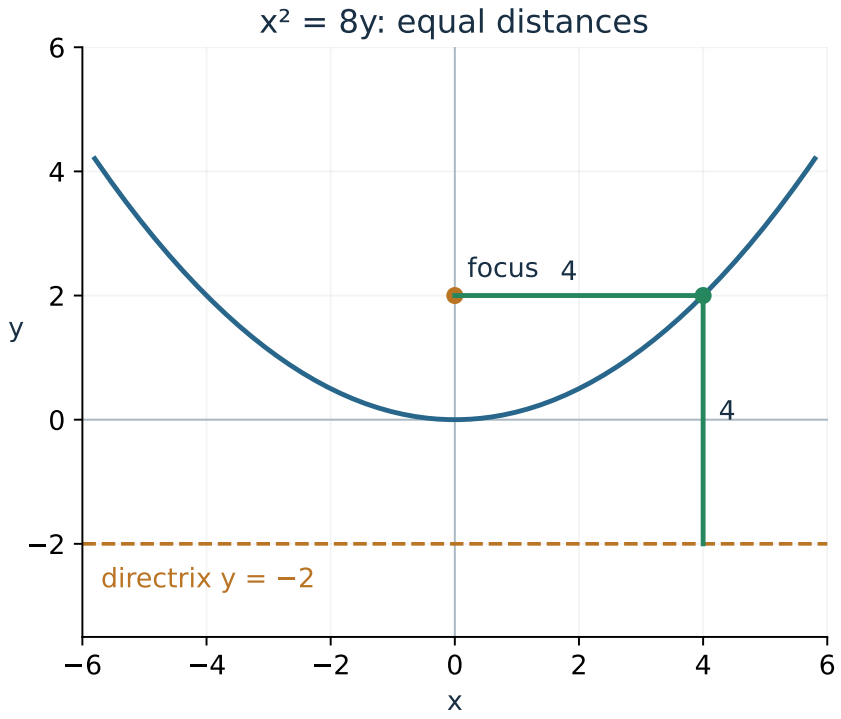


Figure 16: A point on the parabola is equally far from the focus and the directrix.

To derive the equation in the centered case, equate the distances from (x,y) to $(0,p)$ and to $y=-p$:

$$\sqrt{x^2 + (y-p)^2} = |y+p|.$$

Squaring gives $x^2 + (y-p)^2 = (y+p)^2$, which simplifies to $x^2 = 4py$.

For $x^2 = 8y$, $4p = 8$, so $p = 2$. The focus is $(0,2)$, and the directrix is $y = -2$. For $(x-1)^2 = 12(y+2)$, first locate the vertex $(1,-2)$; then $p = 3$, giving focus $(1,1)$ and directrix $y = -5$.

8.7 A hyperbola fixes a difference of distances

A hyperbola consists of points whose absolute difference of distances to two foci is constant. Its two branches lie near the two foci. For the horizontal form

$$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1,$$

the vertices are $(h \pm a, k)$, the foci are $(h \pm c, k)$, and

$$c^2 = a^2 + b^2, \qquad y - k = \pm\frac{b}{a}(x-h)$$

are the two asymptotes.

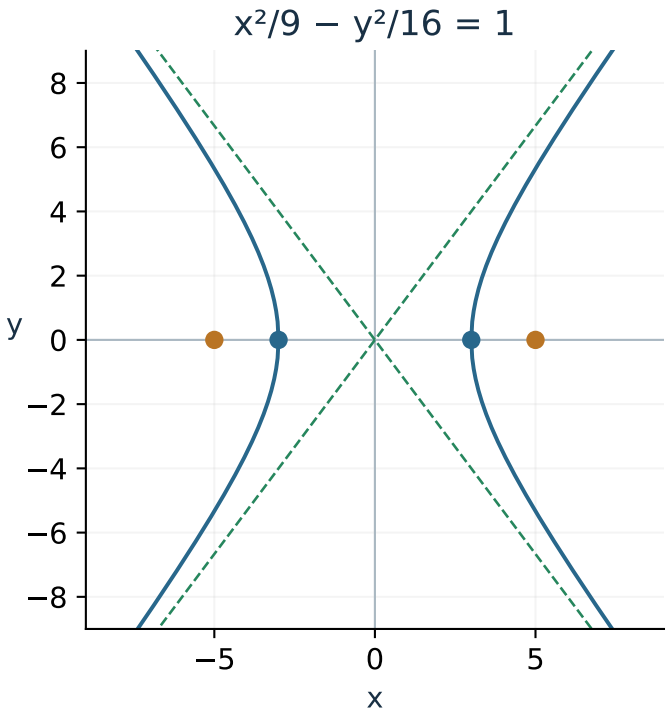


Figure 17: The branches of a hyperbola approach two asymptotes. Its foci lie farther from the center than its vertices.

For $x^2/9 - y^2/16 = 1$, we have $a = 3, b = 4, c = 5$. The vertices are $(\pm 3, 0)$, the foci $(\pm 5, 0)$, and the asymptotes $y = \pm 4x/3$. Far from the center, the constant 1 becomes small relative to the squared terms, explaining why the graph approaches those lines.

The sign difference is important. Ellipses use a sum of positive squared terms and remain bounded. Hyperbolas use a difference and extend along two branches.

8.8 Connect the descriptions

Use the Curves and Conics Explorer to move a point along a selected curve. Its slider records the fraction of the displayed path traversed. For circles and ellipses, one complete traversal corresponds to an angle change of 2π.

A parametric description helps with motion and plotting. A Cartesian equation helps identify centers, axes, and intercepts. A focal definition explains a distance relationship. Translate among these descriptions while keeping the parameter interval and orientation visible.

8.9 Review exercises

1. Eliminate t from $x = t + 2,\ y = 3t - 1$, and describe the motion.
2. Find the Cartesian equation for $x = 4\cos t,\ y = 3\sin t$.
3. Compare $(\cos t, \sin t)$ and $(\cos 3t, \sin 3t)$ on $[0, 2\pi]$.
4. Write the projectile equations for a launch from the origin at 20 m/s and $30°$.
5. Identify center and radius of $(x-2)^2 + (y+1)^2 = 16$.
6. Find the foci of $x^2/36 + y^2/20 = 1$.
7. Find the focus and directrix of $(x-1)^2 = 12(y+2)$.
8. Find the vertices and asymptotes of $x^2/4 - y^2/9 = 1$.
9. Parameterize the circle of radius 3 centered at $(-2, 1)$.
10. Give a parameter interval tracing only the lower half of $x = 4\cos t,\ y = 3\sin t$.

8.10 Check your understanding

The Try it gives $y = (x-7)/2$, with endpoints $(1, -3)$ and $(9, 1)$. The line equation describes a whole line; the parameter interval selects this particular segment and the direction from its first endpoint to its second.

9 Sequences and Series

Suppose a theater adds two seats to each successive row. We can ask how many seats are in row 15, or how many seats are in the first 15 rows altogether. Those questions use the same pattern but ask for different quantities.

A sequence records individual values. A series adds them. We will build formulas from repeated addition and multiplication, keeping the meaning of each index and total in view.

9.1 Separate a term from a running total

A sequence assigns a number a_n to each integer index in a specified list. Usually we begin at $n = 1$, though a time model may begin at $n = 0$. The first terms of $a_n = 3 + 2(n - 1)$ are $3, 5, 7, 9, \ldots$.

The tenth term is a_{10}. The sum of the first ten terms is S_{10}:

$$S_n = a_1 + a_2 + \cdots + a_n.$$

The subscript tells you how far along the list to go. It does not mean multiplication.

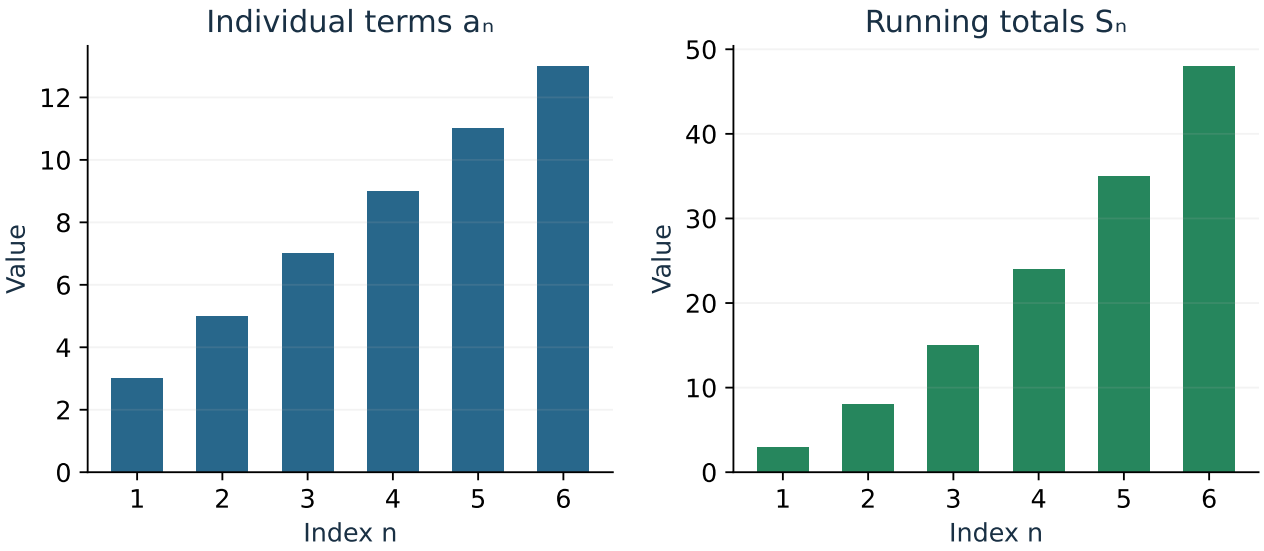


Figure 18: Individual terms of a sequence and the sums accumulated through those terms answer different questions.

For $3, 5, 7, 9, \ldots$, the first four partial sums are $3, 8, 15, 24$. The fourth term is 9, while the fourth partial sum is 24.

Try it. List the first five terms of $a_n = 7 - 3(n - 1)$, then calculate their sum. Some terms are negative; include their signs.

9.2 Repeated addition produces an arithmetic sequence

An **arithmetic sequence** adds the same common difference d at each step:

$$a_{n+1} = a_n + d.$$

Starting from a_1, reaching term n requires $n - 1$ additions. Therefore

$$a_n = a_1 + (n - 1)d.$$

The recursive rule tells you how to take a step; the explicit formula lets you jump directly to any term.

Worked example: theater rows. The first row has 12 seats, and each next row has 2 more. Row 15 has

$$a_{15} = 12 + (15 - 1)(2) = 40$$

seats. There are 14 increases between the first and fifteenth rows.

To add the rows, write the sequence forward and backward:

$$S_n = a_1 + a_2 + \cdots + a_n,$$
$$S_n = a_n + a_{n-1} + \cdots + a_1.$$

Each column adds to $a_1 + a_n$, and there are n columns. Thus

$$S_n = \frac{n}{2}(a_1 + a_n) = \frac{n}{2}(2a_1 + (n - 1)d).$$

For the first 15 rows, the average of the endpoint counts is $(12 + 40)/2 = 26$. Multiplying by 15 gives 390 seats. The formula also works for an odd number of terms: the middle term equals that same average.

9.3 Repeated multiplication produces a geometric sequence

A **geometric sequence** multiplies by the same ratio q at each step:

$$a_{n+1} = qa_n, \qquad a_n = a_1 q^{n-1}.$$

For $a_1 = 80, q = 1/2$, the terms are $80, 40, 20, 10, 5, \ldots$. With $0 < q < 1$, positive terms decrease toward zero. With $q > 1$, positive terms grow. Negative ratios alternate signs.

A percentage change specifies a multiplication factor. An 8% increase multiplies by 1.08; a 12% decrease multiplies by 0.88.

Worked example: define the initial index. A population begins at 500 and grows 8% per generation. If $P_0 = 500$ records the initial population, then

$$P_n = 500(1.08)^n.$$

If the initial observation is labeled $a_1 = 500$, the formula is $a_n = 500(1.08)^{n-1}$. The physical model is the same; the labels differ by one.

9.4 Add a finite geometric pattern

Write a partial sum and its multiple by q:

$$S_n = a_1 + a_1 q + \cdots + a_1 q^{n-1},$$
$$qS_n = \qquad a_1 q + \cdots + a_1 q^{n-1} + a_1 q^n.$$

Subtracting cancels every middle term:

$$(1 - q)S_n = a_1 - a_1 q^n, \qquad S_n = a_1 \frac{1 - q^n}{1 - q} \quad (q \neq 1).$$

For $q = 1$, all terms equal a_1, so the total is na_1.

Worked example: a finite total. For $5, 3, 1.8, \ldots$, the ratio is 0.6. The sum through term 5 is

$$S_5 = 5\frac{1 - 0.6^5}{1 - 0.6} = 11.528.$$

This adds five terms. The fifth term alone is $a_5 = 5(0.6)^4 = 0.648$.

The finite-sum formula works for growing, decreasing, or alternating geometric patterns, provided $q \neq 1$. It does not require convergence.

9.5 Read sigma notation as an instruction to add

The notation

$$\sum_{k=1}^{4}(2k+1)$$

says: substitute $k = 1, 2, 3, 4$, then add the four results. It becomes $3 + 5 + 7 + 9 = 24$.

The letter k is an index used inside the instruction. The bottom and top bounds determine which values to use. For $\sum_{k=0}^{5} 2(3/2)^k$, there are six terms, since both endpoints count.

Before applying a formula, write the first few terms and the last term. This checks the first value, ratio, and number of terms.

9.6 Can infinitely many additions have a finite total?

For the geometric pattern with $|q| < 1$, the power q^n approaches zero. The finite-sum formula then approaches

$$\sum_{n=0}^{\infty} aq^n = \frac{a}{1-q}.$$

An infinite sum means the limit of its partial sums. We calculate finite totals and ask what value they approach.

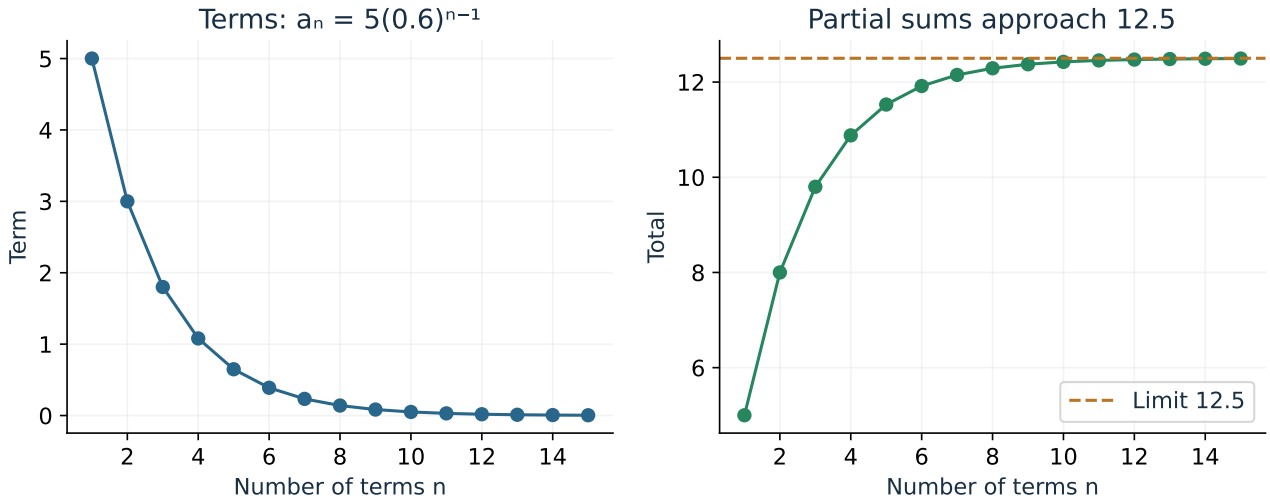


Figure 19: Geometric terms approach zero while their running totals approach a finite value. Ratios with magnitude at least one behave differently for nonzero first terms.

For $5 + 3 + 1.8 + \cdots$, the limiting total is $5/(1 - 0.6) = 12.5$. After n terms, the remaining amount is

$$\frac{5(0.6)^n}{1-0.6}.$$

This also measures how close a partial sum is to the limit.

Worked example: idealized bouncing. Drop a ball from 5 m, with each rebound reaching 60% of the previous height. The first drop contributes 5 m. Each later rebound height contributes twice, once upward and once downward:

$$5 + 2(5)(0.6 + 0.6^2 + \cdots) = 5 + 10\frac{0.6}{1-0.6} = 20 \text{ m}.$$

The model describes infinitely many idealized rebounds with a finite total travel distance.

For a geometric series with nonzero first term, $|q| \geq 1$ prevents convergence. With $q = 2$, terms grow; with $q = -1$, partial sums oscillate. Also, terms approaching zero alone do not guarantee convergence for every kind of series; the conclusion above follows from the geometric formula.

Try it. Find the sum of $4 - 2 + 1 - \frac{1}{2} + \cdots$. Identify its ratio and explain why the alternating signs are allowed.

9.7 A feedback rule can hide a geometric pattern

Consider

$$a_{n+1} = \frac{a_n + 6}{2}, \qquad a_1 = 0.$$

Each step averages the current value with 6. The first terms are $0, 3, 4.5, 5.25, 5.625, \ldots$. The changes are not constant, and neither are the ratios of consecutive terms.

Find the equilibrium by asking which value would remain fixed:

$$L = \frac{L + 6}{2} \quad \Longrightarrow \quad L = 6.$$

Now examine the difference $e_n = a_n - 6$. Subtracting 6 from the recurrence gives

$$e_{n+1} = \frac{e_n}{2}.$$

The **difference from equilibrium** is geometric, even though the original sequence is not. Since $e_1 = -6$,

$$a_n = 6 - 6\left(\frac{1}{2}\right)^{n-1}.$$

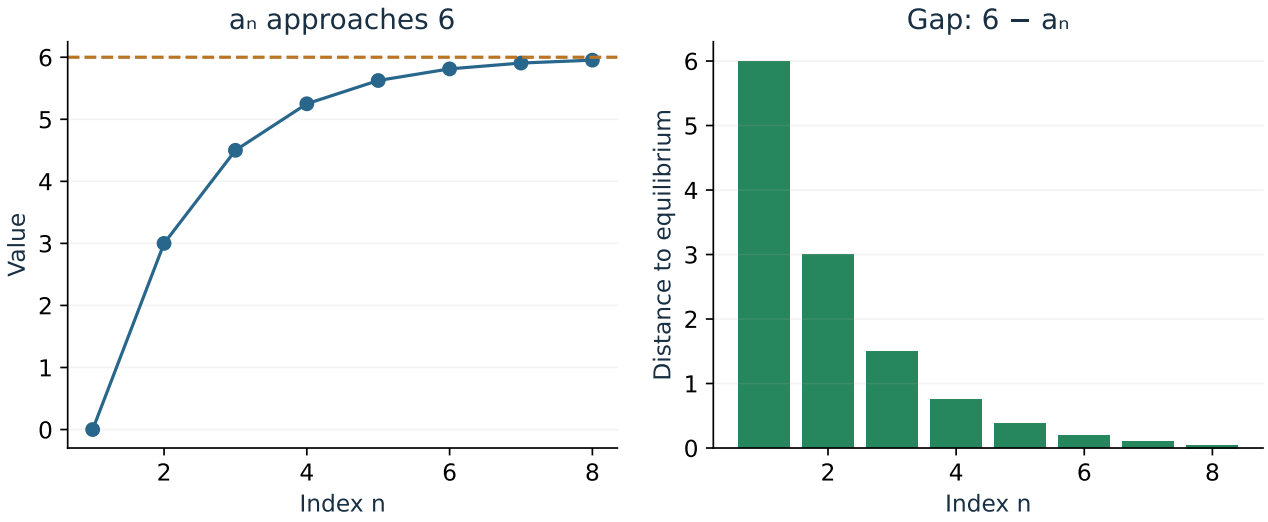


Figure 20: A feedback sequence approaches its equilibrium while the remaining gap halves at every step.

Use the Sequence Explorer to compare terms and partial sums of arithmetic and geometric sequences. For the feedback example, compute the terms by hand and use the formula for the remaining gap to explain the approach.

9.8 Review exercises

1. List five terms of $a_n = 7 - 3(n-1)$, and find a_{20}.
2. For $a_1 = 4, d = 5$, calculate S_{30}.
3. For $a_1 = 81, q = 1/3$, find a_6.
4. Compute $\sum_{k=0}^{5} 2(3/2)^k$ exactly.
5. Explain the difference between $\sum_{k=1}^{n} k$ and its final term n.
6. Find the sum of $5 + 5/2 + 5/4 + \cdots$.
7. A machine initially worth \$12,000 loses 12% of its value each year. Write its value after n years.
8. For $a_{n+1} = (a_n + 10)/2,\ a_1 = 2$, find the equilibrium and an explicit formula.
9. A theater has 20 rows, with 14 seats in the first and 3 more in each next row. Find the total.
10. Explain why $1 - 1 + 1 - 1 + \cdots$ has no limiting partial sum.

9.9 Check your understanding

The first Try it gives $7, 4, 1, -2, -5$, with sum 5. The alternating geometric series has ratio $-1/2$ and sum $4/(1 + 1/2) = 8/3$. Its partial sums approach the limit from alternating sides.

10 Synthesis: Motion and Patterns

Imagine a light attached to a rotating wheel. We can record its angle, plot its coordinates, describe its position as a complex number, or take a photograph every few seconds. Each description records a different aspect of one motion.

This chapter brings those descriptions together. We will first follow a continuous rotation, then sample it, and finally combine repeated rotation with shrinking or growing distances. The goal is to choose and connect the tools developed throughout the course.

10.1 Begin with a physical description

A point moves counterclockwise on a circle of radius 5 m, completing one turn every 20 s. It begins at the rightmost point.

The angular speed is one full turn divided by the time per turn:

$$\omega = \frac{2\pi}{20} = \frac{\pi}{10} \text{ rad/s}.$$

At time t, the angle is $\theta(t) = \pi t/10$. If the point had begun at another angle θ_0, we would add that initial angle.

The unit-circle formulas give the position:

$$x(t) = 5\cos\left(\frac{\pi t}{10}\right), \qquad y(t) = 5\sin\left(\frac{\pi t}{10}\right).$$

Polar coordinates record $(r, \theta) = (5, \pi t/10)$, while complex form records

$$z(t) = 5\left(\cos\frac{\pi t}{10} + i\sin\frac{\pi t}{10}\right).$$

All three describe a position 5 m from the origin at the same angle.

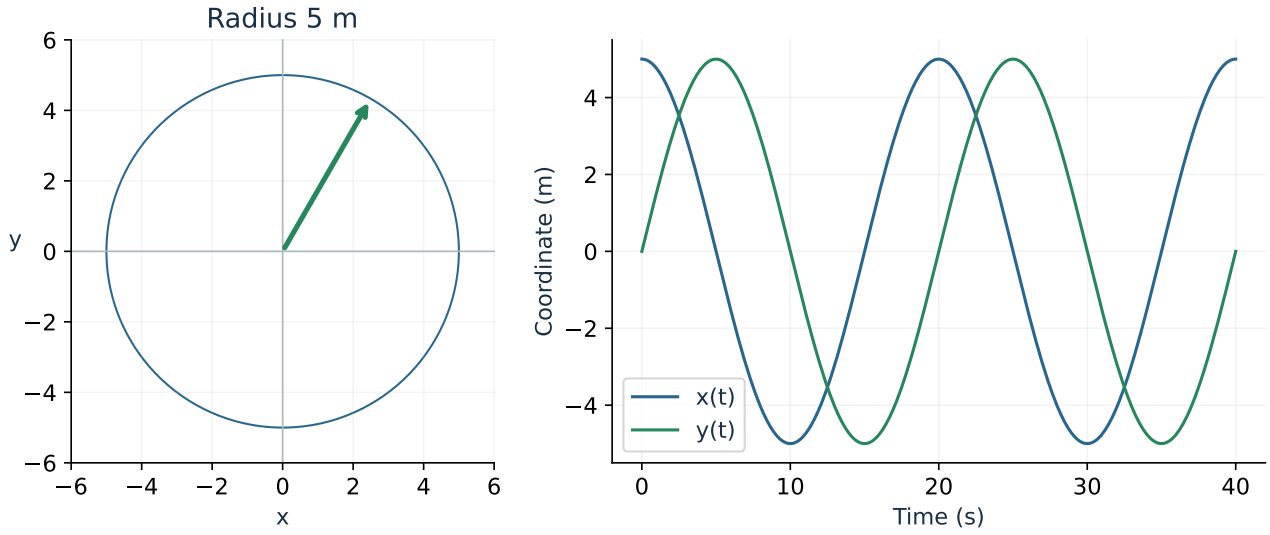


Figure 21: A point on a circle produces horizontal and vertical coordinate waves with the same period.

The coordinate waves have amplitude 5 and period 20 s. The circle itself has equation $x^2 + y^2 = 25$. That equation records the path; the time formulas additionally record orientation and timing.

10.2 Compare position, displacement, and distance traveled

After 5 seconds, the point moves from $(5, 0)$ to $(0, 5)$. Its displacement is

$$\Delta \mathbf{r} = \langle 0 - 5, 5 - 0 \rangle = \langle -5, 5 \rangle,$$

with magnitude $5\sqrt{2}$ m.

The distance traveled is a quarter of the circumference:

$$s = 5\left(\frac{\pi}{2}\right) = \frac{5\pi}{2} \text{ m}.$$

The curved distance exceeds the straight displacement magnitude. The average velocity over this interval is the displacement divided by elapsed time, $\langle -1, 1 \rangle$ m/s. The average speed is distance divided by time, $\pi/2$ m/s.

For constant angular speed, speed along the circle is $R|\omega|$. The instantaneous direction is tangent to the circle. We can recognize that geometric direction without differentiating the coordinate formulas.

Try it. For this same motion, compare displacement and distance traveled over a complete 20-second turn.

10.3 Sampling turns continuous motion into a sequence

Suppose a camera records the position every 5 seconds. The angle advances by

$$\delta = \omega \Delta t = \frac{\pi}{10}(5) = \frac{\pi}{2}$$

between photographs. Beginning with index $n = 0$, the first five positions are

$$z_0 = 5, \quad z_1 = 5i, \quad z_2 = -5, \quad z_3 = -5i, \quad z_4 = 5.$$

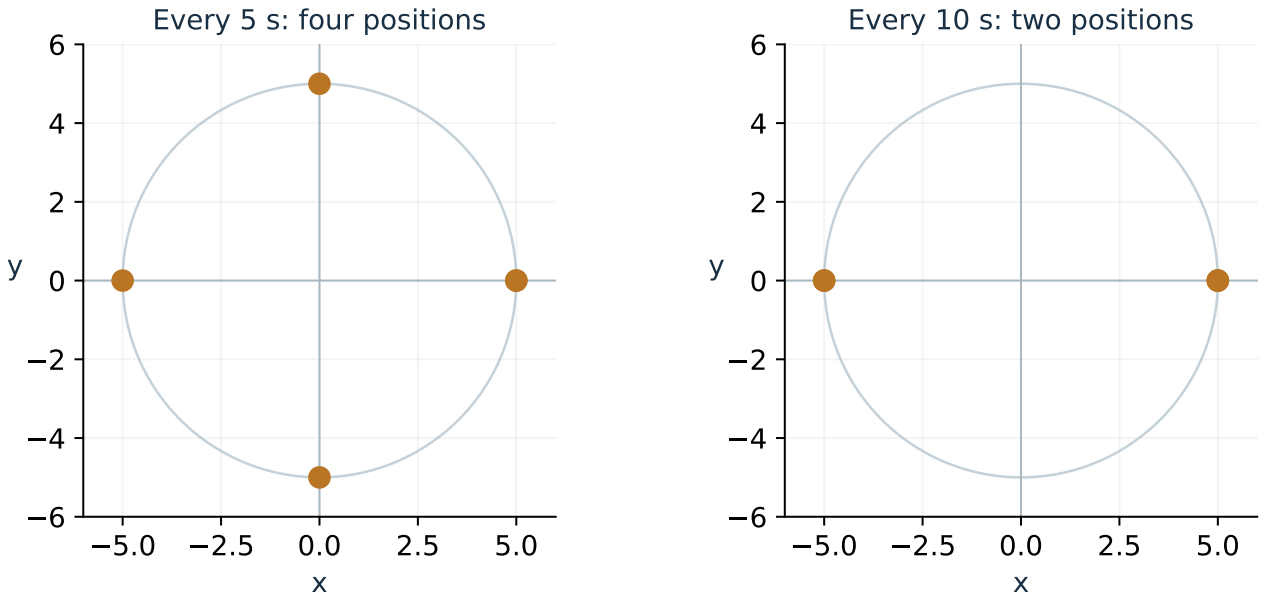


Figure 22: Sampling every quarter-turn yields four distinct positions; other sampling intervals can reveal different patterns of the same motion.

Each sample is the previous one multiplied by i. We have both a recurrence and an explicit formula:

$$z_{n+1} = iz_n, \qquad z_n = 5i^n.$$

De Moivre's theorem expresses the same result as

$$z_n = 5\big(\cos(n\pi/2) + i\sin(n\pi/2)\big).$$

More generally, a fixed angle step δ corresponds to

$$u = \cos\delta + i\sin\delta, \qquad z_{n+1} = uz_n.$$

Repeated multiplication gives $z_n = u^n z_0$.

The sampling interval matters. Photographing the original wheel every 20 seconds would always capture the same point. Those photographs alone would conceal the motion between them. Sampling every 10 seconds gives two alternating positions.

10.4 Decide whether sampled positions repeat

For a nonzero starting point and unit-length multiplier, a sample returns to its start when the accumulated angle is a whole number of turns:

$$n\delta = 2\pi k$$

for integers $n > 0, k$.

If $\delta/(2\pi) = p/q$ in lowest terms with $q > 0$, the first return is after q steps. For example, $\delta = 4\pi/5$ is $2/5$ of a turn, so five steps close the sampled orbit. The points are visited in a star order rather than around consecutive edges.

If $\delta/(2\pi)$ is irrational, no positive integer number of steps gives an exact return. A finite computer display can look almost closed, so distinguish a close approach from mathematical equality.

10.5 Add a scale change at each step

Now let the multiplier have modulus $\rho > 0$:

$$q = \rho(\cos\delta + i\sin\delta), \qquad z_{n+1} = qz_n.$$

After n steps,

$$z_n = q^n z_0, \qquad |z_n| = \rho^n |z_0|.$$

The angle advances by $n\delta$, measured modulo full turns. The distance from the origin follows a geometric sequence.

For $0 < \rho < 1$, radii shrink toward zero. For $\rho = 1$, they stay constant. For $\rho > 1$, they grow. These statements describe a nonzero starting point. The origin stays fixed.

Worked example: track three shrinking quarter-turns. Start at $z_0 = 4$ and multiply by $q = i/2$:

$$z_1 = 2i, \qquad z_2 = -1, \qquad z_3 = -\frac{1}{2}i.$$

The radius halves each step: $4, 2, 1, 1/2$. The angle adds $90°$: $0°, 90°, 180°, 270°$. The explicit formula is

$$z_n = 4\left(\frac{1}{2}\right)^n \big(\cos(n\pi/2) + i\sin(n\pi/2)\big).$$

The angular pattern repeats, while the positions do not: their nonzero magnitudes keep decreasing.

Try it. Begin with $z_0 = 2$ and use $q = \frac{1}{2}(\cos 60° + i\sin 60°)$. Find the modulus, direction, and rectangular value of z_3.

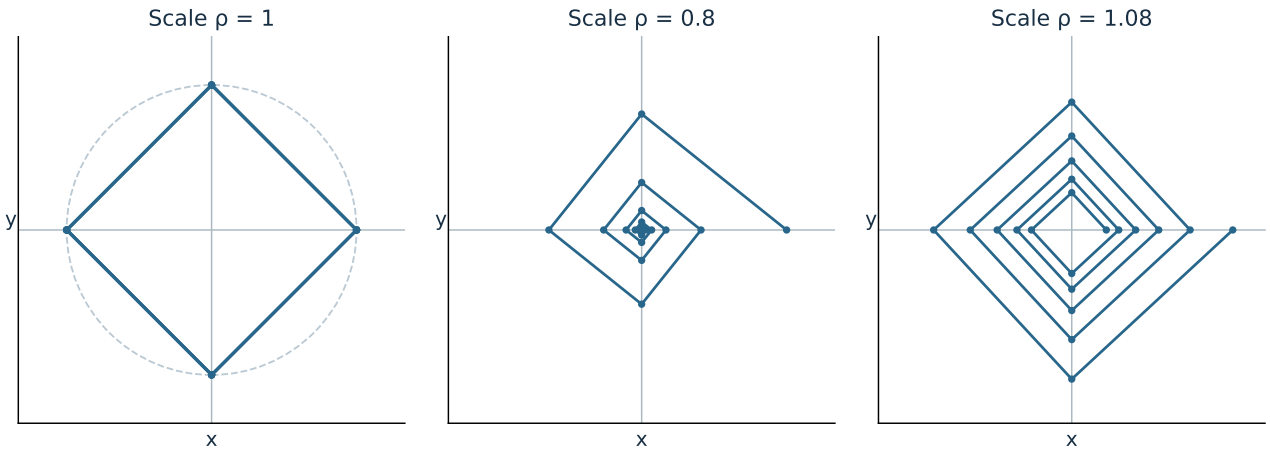


Figure 23: A repeated quarter-turn stays on a circle when the scale is one, spirals inward for a scale below one, and outward for a scale above one.

10.6 Adding step vectors builds a different sequence

The recurrence above multiplies the **position** at each step. Another model specifies the **displacement** during each step. If those steps are $\Delta z_n = aq^n$, starting from position z_0, then

$$z_N - z_0 = \sum_{n=0}^{N-1} aq^n = a\frac{1-q^N}{1-q} \qquad (q \neq 1).$$

The geometric-sum algebra also works for complex numbers, since multiplication and subtraction obey the same distributive rules.

For steps $1, i, i^2, i^3$, the total displacement is $1+i-1-i = 0$. From the origin, the running positions are $0, 1, 1+i, i, 0$. Those positions are the corners of a square.

For shrinking steps $a(i/2)^n$, the step lengths approach zero. Their total displacement approaches $a/(1-i/2)$. The final position depends on both that sum and the starting point. Always decide whether a formula describes locations or movements between locations.

10.7 Choose a representation that answers the question

A model begins with quantities and units. For rotating machinery, angle and period may be the natural measurements. For navigation, components help add movements. For repeated scaling and rotation, complex multiplication gives a compact rule. For a path with a particular timing, parametric equations supply both coordinates at each instant.

These choices also reveal the assumptions. A constant-speed circle needs a fixed radius and angular speed. An ellipse has a different relation between its horizontal and vertical coordinates. An idealized projectile uses constant horizontal velocity and downward acceleration. Select a model because its assumptions fit the question.

Use the Motion Synthesizer to vary the scale and angle independently. Fix one control while investigating the other. Then explain the plotted pattern through its radius sequence and angle sequence.

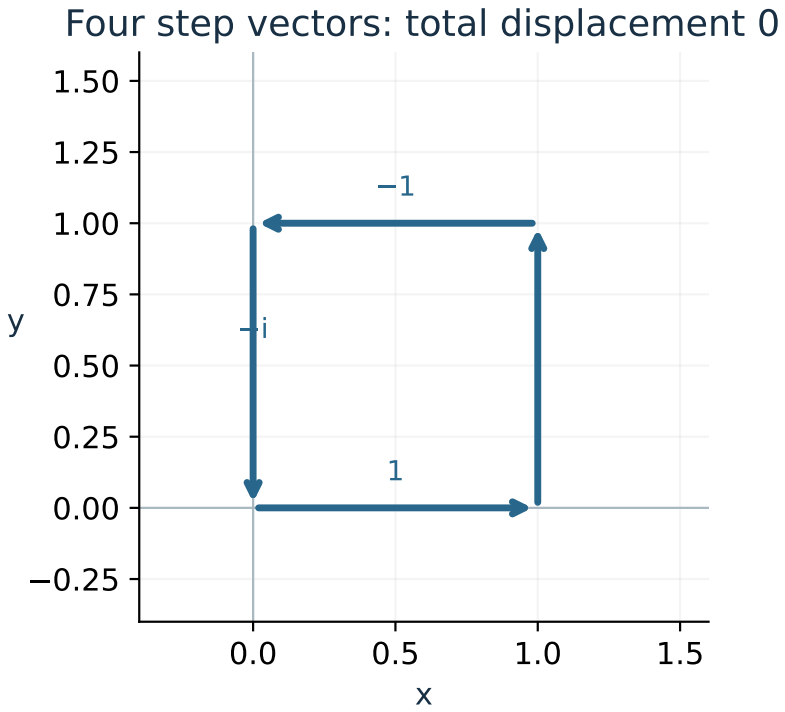


Figure 24: Adding four quarter-turn step vectors makes a closed polygon. The step vectors are different from the running positions.

10.8 Review exercises

1. Write Cartesian, polar, and complex formulas for radius 5 and period 20 s, starting on the positive horizontal axis.
2. Sample every 5 s. Give the first five complex positions and the repetition period in samples.
3. For $z_{n+1} = iz_n,\ z_0 = 1 + i$, find $z_1, \ldots, z_4$.
4. For $z_{n+1} = (i/2)z_n,\ z_0 = 4$, find z_4 and describe its magnitude and direction.
5. Eliminate t from $x = 3\cos t,\ y = 2\sin t$.
6. Explain why a circle can repeat while a shrinking spiral from a nonzero point does not.
7. Interpret $1 + i + i^2 + i^3$ as four step vectors.
8. Describe one process with at least three mathematical representations from the course.
9. A circle is sampled with angle step $4\pi/5$. How many steps give its first return?
10. Explain what photographs every full revolution would reveal and what information they would miss.

10.9 Check your understanding

Over a complete turn, displacement is zero and traveled distance is 10π m. In the shrinking $60°$ example, after three steps the modulus is $2(1/2)^3 = 1/4$, the direction is $180°$, and $z_3 = -1/4$.